Module 4: Traffic Engineering (8 Lectures)

- Traffic Characteristics, Speed, Journey Time and Delays, Vehicle Volume Counts, Origin and Destination Studies, Analysis and Interpretation of Survey Data, Traffic Operations
- Design of Signals and Rotary intersections, Parking Space Design, Highway Lighting, Planning and Administration, Road Markings, Signs

Road Accidents and Safety: Classification, Causes, Mitigation and Control Measures, Aspects of Safety in Usage of Roads, Type and Design of anti-crash barriers

- Introduction to Intelligent Transport Systems (ITS).

Traffic Engineering:

1) Traffic Engg. which deals with planning & geometric design of roads on one hand & **regulation and control** on other hand.
2) Traffic Engineering which deals with the **improvement of traffic performance** on road network and terminals through **systematic traffic studies, scientific analysis & engineering applications.**
3) Traffic Engg. is the science of measuring traffic & travel, the study of basic laws relating to traffic flow & generation & application of this knowledge to the professional practice of planning, designing & operating traffic systems to achieve safe & efficient movements of persons & goods.

Objectives of Traffic Engg.:

1) To provide efficient flow of traffic.
2) To provide free flow of traffic.
3) To provide rapid flow of traffic.
4) To provide safety to traffic.

Scope of Traffic Engineering:

The basic objective of traffic engineering is to achieve efficient, free & rapid flow of traffic, with **least number of traffic accidents.**

1) Traffic Characteristics
2) Traffic Studies and analysis
3) Planning analysis
4) Geometric design
5) Traffic operation, regulation & control.
6) Road safety aspects
7) Administration & management

3 E’s of Traffic Engineering:

Phases of Traffic Engineering are implemented with the help of Engineering, Enforcement & Education is known as 3 E’s.

1) **Engineering**
2) **Enforcement**
3) **Education**

- **Engineering deals with improvement of road geometrics, providing road facilities & installation of suitable traffic control devices.**
- **Enforcement is made through traffic laws, regulation & control.**
- **Education may be imparted by sufficient publicity and through television.**

1) **Engineering:** Engineering deals with improvement of road geometrics, providing road facilities & installation of suitable traffic control devices. The geometric design features of the road such as sight distance, width of pavements, width of shoulders, horizontal & vertical alignment design details & intersection design elements are checked & corrected if necessary.

2) **Education:** Education may be imparted by sufficient publicity and through television, poster, newspaper, etc. Education of road users (pedestrian, passenger, driver, students, etc.) & Safety drives (Training courses, workshop, traffic safety week, etc).

3) **Enforcement:** Enforcement is made through traffic laws, regulation & control. The various measures of enforcement that may be useful to prevent accidents at spot prone to accidents are enumerated here. The motor vehicle rules are revised from time to time to make them more comprehensive. Speed control, Traffic control devices, Training & supervision, Medical check, special precaution for commercial vehicle, etc.

Traffic Characteristics

Basic Traffic Characteristics in transport engineering are an important aspect that covers two major characteristics in the study of transportation engineering.

- Traffic characteristics are **necessary for road design**.
- Traffic characteristics also **help in the design of vehicles.**

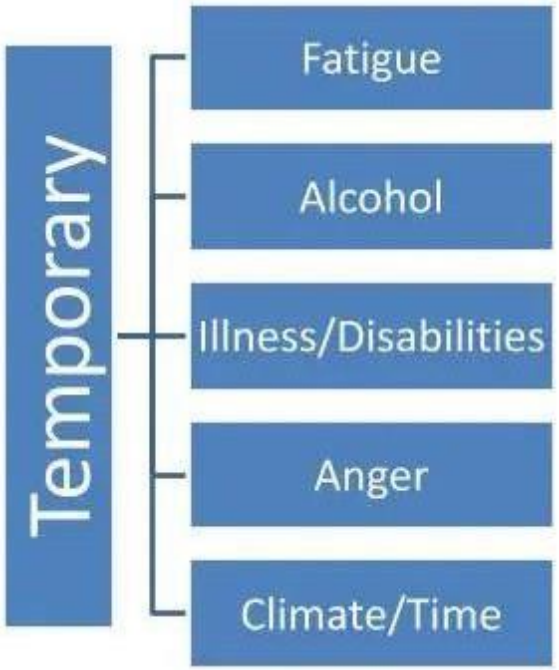
- In road design, traffic characteristics help in deciding speed limit, stopping sight distance, overtaking sight distance, design of traffic islands, speed of the vehicle, design of parking, right of way, etc.
- The weight of vehicles, the height of vehicles, and other aspects of vehicle design too require an understanding of traffic characteristics.
- Placing signs and symbols on the road also needs knowledge of road user characteristics. The color of these signs, symbols, and other elements of the road are decided after considering traffic characteristics.

Types of Traffic Characteristics in Transport Engineering

There are two main types of basic traffic characteristics as follows:

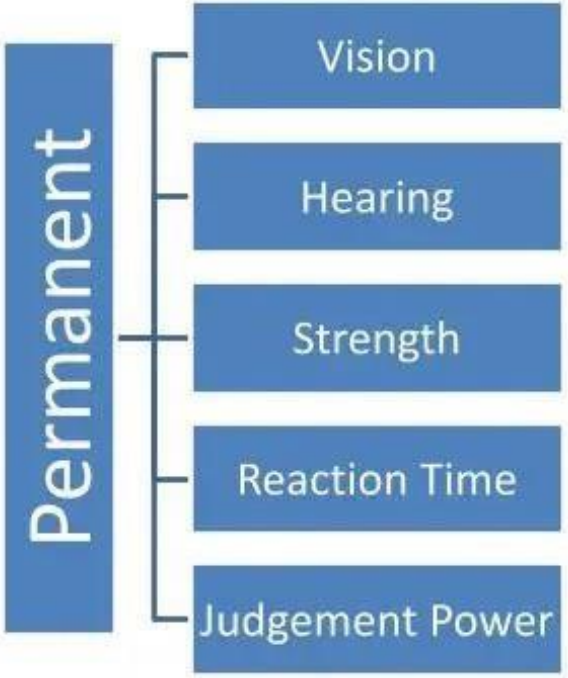
A. Road user characteristics

B. Vehicular characteristics



A.Road user characteristics

1. **Internal road user characteristics** are the internally related characteristics of road users (human beings). These are again of two types:



- Temporary internal road user characteristics
- Permanent road user characteristics

Fatigue

- Fatigue causes:
- Increased reaction time
- Ineffective judgment power
- Drowsiness
- All the above factors can contribute to accidents.

Alcohol

- Alcohol affects a person both physically and mentally.
- Alcohol causes:
- Reduced attention span
- Increased reaction time
- Ineffective judgment power
- Loss of control over one's own body

Illness/Disabilities

- This does not affect directly, but a person with a disability tends to have more emotional reactions.

Anger

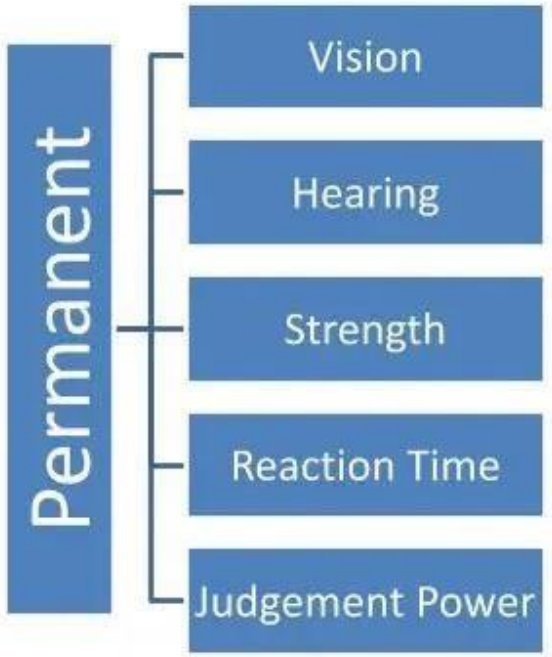
- Anger develops rigid attitudes which can affect the road user negatively.

Climate, season, and time

- The above factors affect the judgment of a road user.

1. Vision

- **Eye sight**: The ability to perceive objects in average illumination falls under eye sight.
- **Peripheral vision**: Though not very clear, objects are visible beyond the cone of clear vision. This is peripheral vision and the cone of peripheral vision in horizontal degree is up to 160 º.
- **Eye movement**: While traversing on road, the road user has to shift his eyes for scanning.
- **Colour vision**: Colour helps in the clear perception of objects.
- **Glare vision and recovery**: Extremely bright light that blinds the vision is glare.
- **Perception of time and space**: This factor becomes important in overtaking operations.



2. Hearing Hearing is not a significant factor in road design.

3. Strength

It is required in vehicle maneuvering operations.

4. Reaction time

It is the time Lapsed In understanding and adapting to a traffic situation.

5. Judgement power

Judgment power is how a person reacts in a specific situation.

2. External Road User Characteristics

Environmental factors are included in external road user characteristics such as:

- facilities provided for traffic
- atmospheric conditions
- locality
- land use
- altitude

B. Vehicular Characteristics

Design of dimensions and weight of vehicles is governed by vehicular characteristics.

Broadly, two types of vehicular characteristics are there –

Category	Characteristic	Description	Relevant IRC Reference
Static Vehicular Characteristics	**Dimensions (Length, Width, Height)**	These physical dimensions affect lane width, turning radii, clearances, and parking design.	IRC:86 (Geometric Design Standards)
	Gross Vehicle Weight (GVW)	The total permissible weight, including vehicle, passengers, and cargo. Important for pavement and bridge design.	IRC:6 (Bridge Design), IRC:37 (Pavement Design)
Dynamic Vehicular Characteristics	**Speed of Vehicle (Design Speed)**	Impacts sight distance, superelevation, curve radii, shoulder width, and vertical gradient. Crucial for geometric road design.	IRC:73 (Geometric Design of Rural Roads)
	Braking System of Vehicle	Affects stopping sight distance (SSD) and headway. Design must consider reaction time and friction.	IRC:66 (SSD & OSD Guidelines)
	Acceleration of Vehicle	Used in intersection design, merging/diverging sections, and signal timing. Affects capacity and safety of traffic elements.	IRC:106 (Intersection Design)

Types of Speeds as per IRC

Speed Type	Definition	Description / Use
Spot Speed	The speed of a vehicle **at a particular instant or location.**	Used for speed studies, design of speed limits, signal timing, and geometric design checks.
Running Speed	The **average speed maintained by a vehicle while the vehicle is in motion**, excluding stoppage time.	Higher than journey speed. Important in understanding actual travel efficiency.
Journey Speed	The **average speed including all delays** (stops due to signals, traffic, etc.) over the journey.	Measures overall travel time efficiency. Used in transport planning and travel time studies.
Time Mean Speed	The **arithmetic mean of speeds of all vehicles passing a point** during a specified period of time.	Used when spot speed is measured using radar, stopwatch, or sensors at a point.
Space Mean Speed	The **harmonic mean of speeds of all vehicles over a certain length of road.**	More accurate for traffic flow analysis. Used in calculating travel time over a road segment.

Journey Time and Delays

Journey Time

- Travel time is the time taken by a vehicle to traverse a given section of a highway.
- Running time is the time a vehicle is actually in motion while traversing a given section of a highway.
- Delay is the time lost by a vehicle due to causes beyond the control of the driver.

Speed And Delay Survey

- This survey evaluates the quality of the traffic movement along a route and identifies the location, cause and extent of the delays in the same route.
- This allows the planners to develop improvements. Also **it can be used as a tool to compare the pre and post effects of any improvement.**

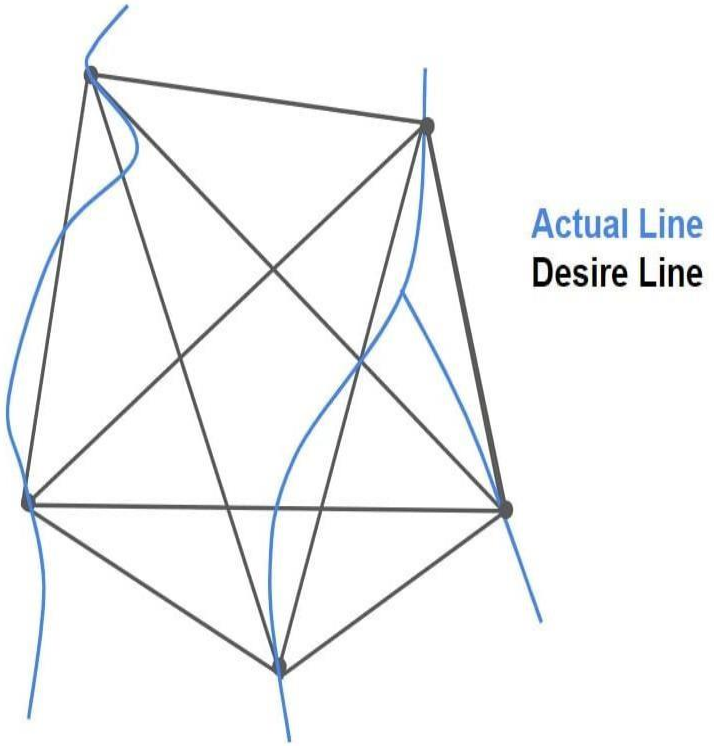
Origin and Destination Studies

Origin and Destination Study (OD Study) is to determine the travel pattern of an area/city.

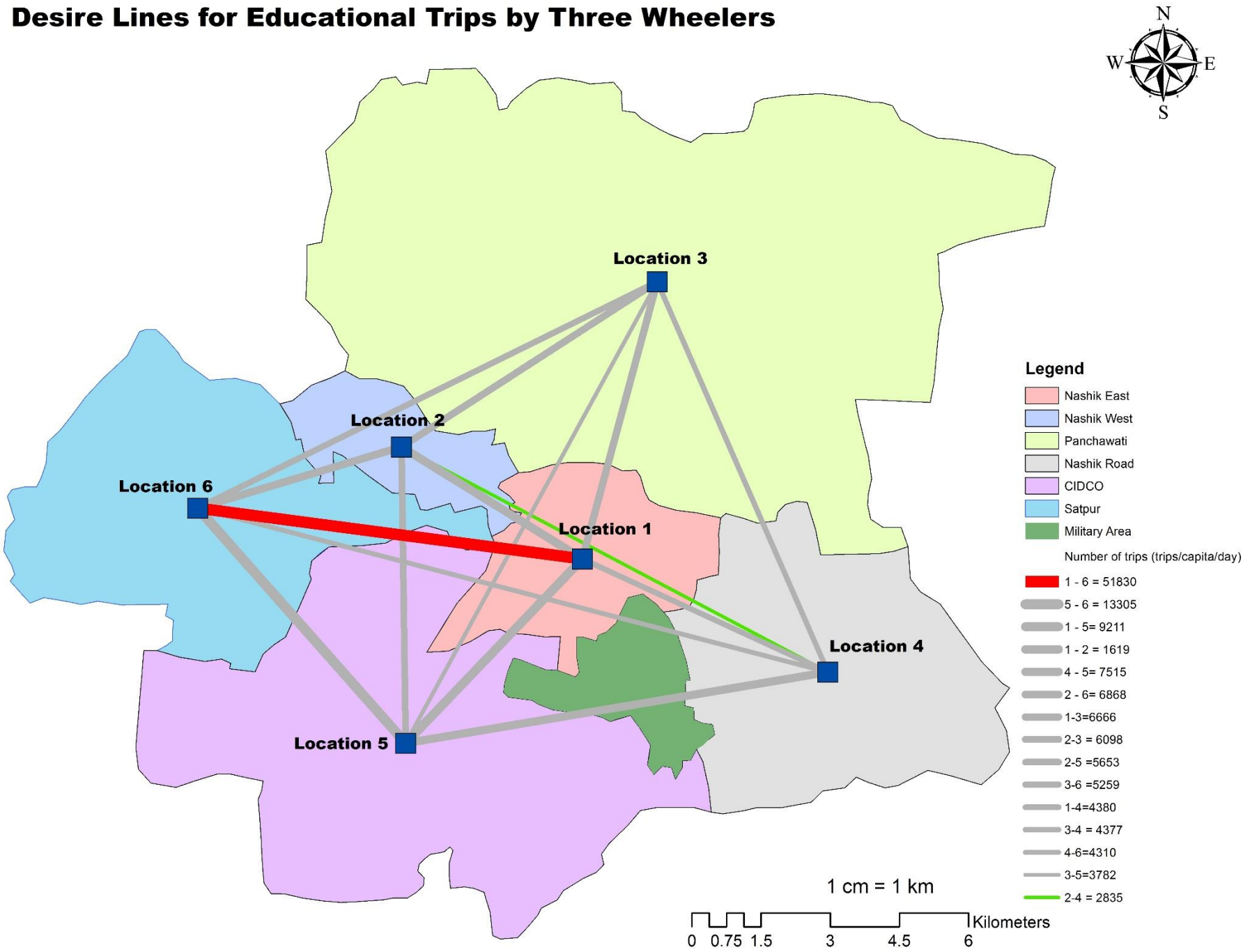
- It makes use **of desire lines of varying thicknesses**.
- O stands from Origin (where you start the travel from) and D stands for destination (where the travel ends).
- If you starting from home, and reach your office - Home is your origin and Office is your destination.
- OD study is to **determine the travel pattern of an area/city.**
- We can find out the origin (start point of travel) and destination (end point of travel) for each person in the city.

Origin and Destination Studies

- This will help in **analysis of the travel pattern observed in the city.**
- The travel pattern is **determined by desire lines which shows us the number of people going from one origin to another destination in the form of lines with varying thickness.**
- From OD study, we can understand the travel pattern of the city.
- Obviously, the thicker the line, more people are travelling along that particular corridor - probably the metro can be planned along that route and attract a lot of people.



Desire Lines for Educational Trips by Three Wheelers



Types of Origin and Destination Studies

Type	Description	Use Cases
1. Roadside Interview Surveys	Direct interviews with drivers at strategically located checkpoints.	Highways, borders, or screen lines.
2. Mail-back / Questionnaire Surveys	Drivers or residents fill out travel details and mail them back.	Urban studies; good for larger sample size.
3. License Plate Matching	Observers record vehicle plate numbers at multiple locations and match them to determine paths.	Low intrusion method for through trips.
4. Home Interview Surveys	Surveys conducted at people's homes to collect detailed trip data.	Comprehensive, but expensive and time-consuming.
5. Telephone / Online Surveys	Contacting respondents via phone or digital platforms to collect O-D data.	Cost-effective; rising in popularity.
6. GPS / Mobile Data Tracking	Using location data from smartphones, GPS, or apps to track trip patterns.	High accuracy, good for large-scale analysis.
7. Bluetooth / Wi-Fi Tracking	Sensors detect MAC addresses of Bluetooth or Wi-Fi devices at various points.	Anonymous, effective in urban corridors.
8. Public Transport Surveys	Surveys on buses/trains to track passenger boarding and alighting locations.	Useful for transit planning.

Applications of O-D Studies

Application	Explanation
1.Traffic Management	Helps identify congested routes and implement control measures (e.g., signal optimization, one-way streets).
2.Road Network Planning	Determines where new roads or highway links are needed by identifying travel demand.
3.Mass Transit Planning	Assists in routing and scheduling of buses and railways by understanding passenger flow.
4. Bypass and Ring Road Planning	If a large number of trips are passing through but not destined within the area, planners can design bypasses.
5. Economic and Land Use Planning	Travel patterns help analyze and forecast land use development, such as shopping centers or industrial zones.
6. Parking Demand Estimation	Data on trip ends helps in planning the number and location of parking spaces.
7. Environmental Impact Assessment	Understanding traffic movement is vital for pollution analysis and mitigation strategies.
8. Freight and Logistics Planning	Essential for planning truck routes, freight terminals, and goods movement optimization.

Parking Studies

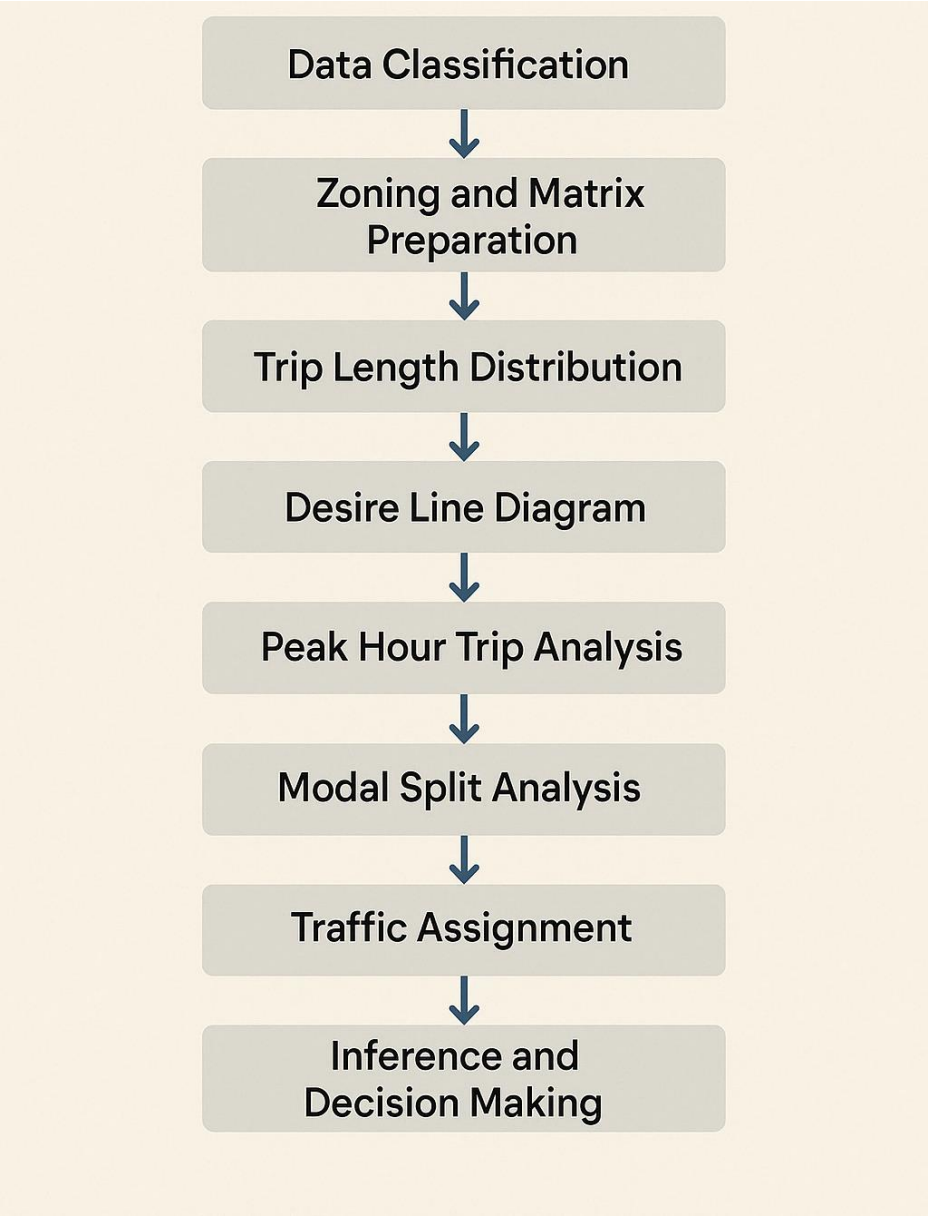
Aspect	Details
Purpose	**Assess parking demand, supply, and traffic impact.**
Types of Surveys	- In-Out Survey: Entry/Exit count - Fixed Period Sampling - License Plate Survey - Cordon Count
Key Parameters	- Parking Accumulation: Vehicles parked at a time - Parking Load: Vehicle-minutes - Turnover: Vehicles per space - Efficiency: Actual vs possible load
Data Collection Tools	Manual count, CCTV, sensors, plate scanners
Applications	- Parking facility planning - Traffic flow improvement - Urban space utilization

ACCIDENT STUDIES & ANALYSIS

S.No.	Component	Description / Standard
1	Need for Study	**Identify accident-prone spots, causes, and mitigation measures**
2	Types of Data Collection	- Police Records - Hospital Records - Spot Investigation - Road Safety Audit
3	Types of Accident Studies	- Spot Studies: At location - Condition Diagram: Road, traffic features - Collision Diagram: Movement & crash pattern
4	Key Accident Parameters	- **Time, Location, Vehicle type, Road & weather conditions, Driver info, Severity**
5	Accident Rate (IRC Formula)	- Per vehicle-km = (No. of Accidents × 10^6) / Vehicle-km - Per road length = Accidents/km/year
6	Severity Index	Severity Index = (Fatal + Serious × Weight) / Total Accidents
7	Statistical Analysis	- Before-After Studies - Comparison of sites - Chi-Square Test
8	Preventive Measures	- Engineering: Signage, geometry - Enforcement: Speed control, fines - Education: Public awareness

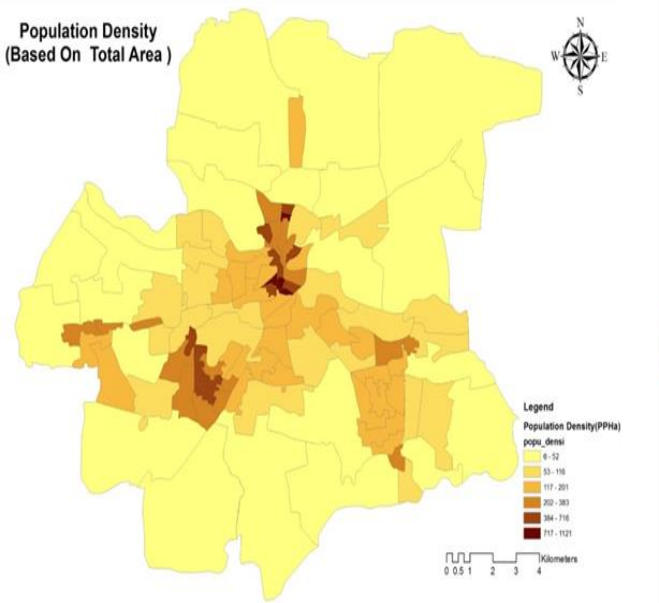
Analysis and Interpretation of Survey Data

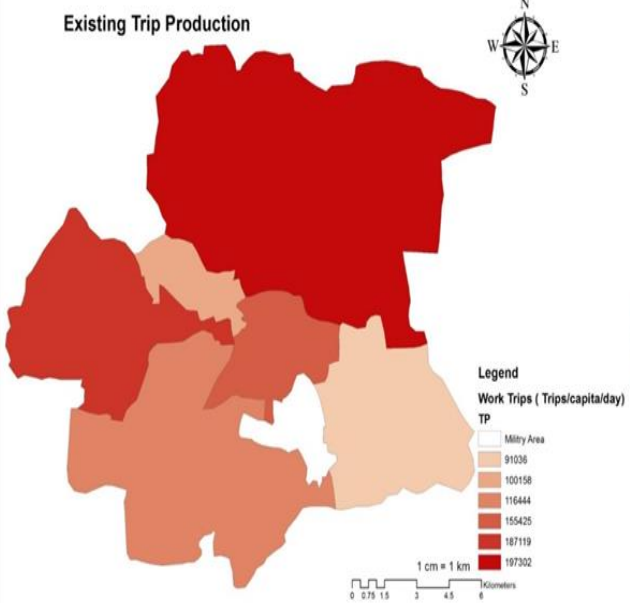
- Traffic surveys consist of various types of data collection techniques done by surveying various aspects of traffic flow.
- These include the following services:
  - Pedestrian counts,
  - Turning movement counts,
  - Bicycle counts,
  - Turning movement counts,
  - Link flow traffic counts,
  - Rail and bus punctuality surveys,
  - Travel time surveys,
  - Roundabout counts,

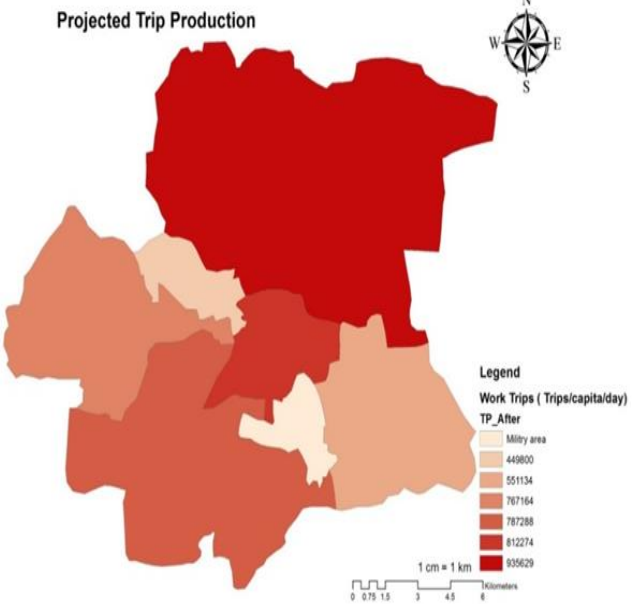


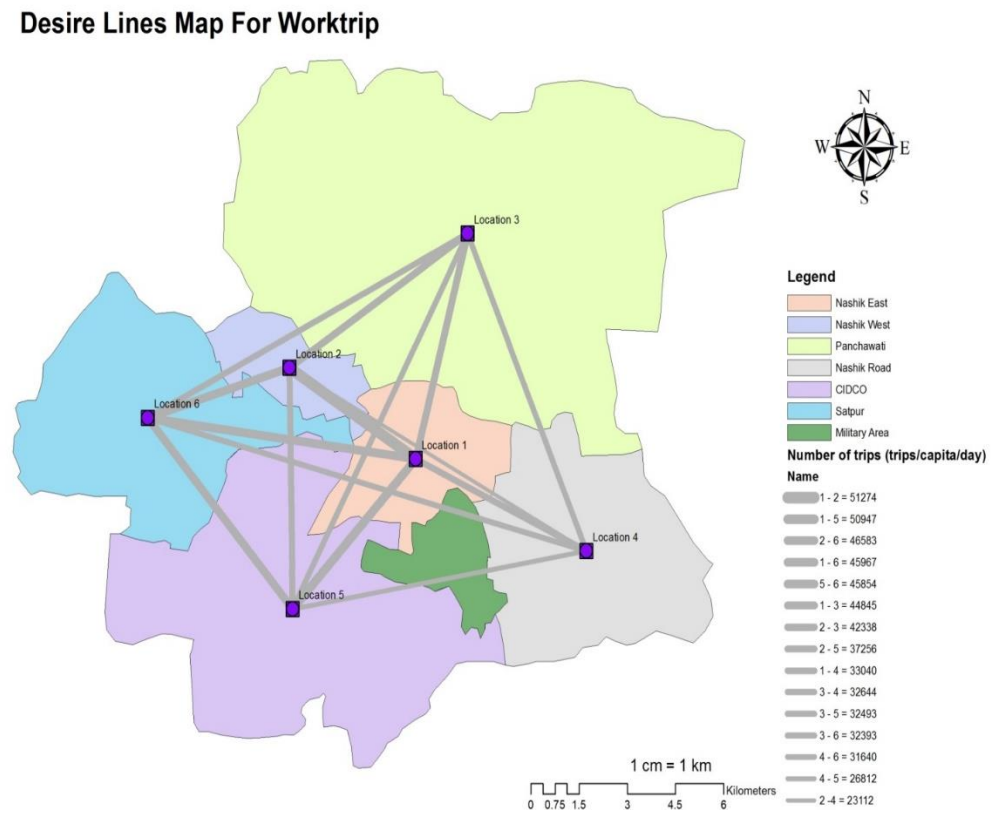
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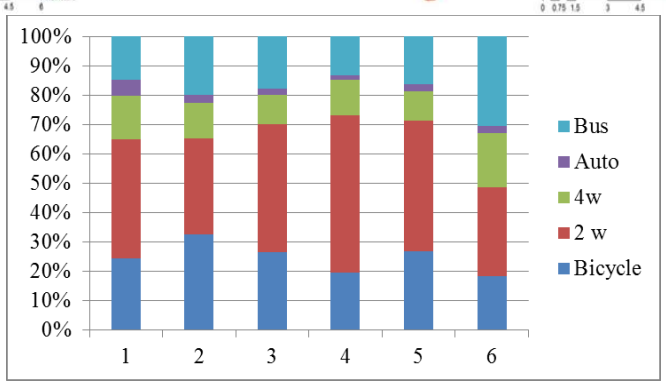
Step	Description
1. Data Classification	Data is sorted based on trip purpose (work, education, shopping), mode of transport (car, bus, etc.), time of travel, and vehicle type.
2. Zoning and Matrix Preparation	The study area is divided into zones, and an O-D matrix is prepared showing the number of trips from each origin to each destination.
3. Trip Length Distribution	Histogram or frequency distribution is prepared showing the number of trips vs. trip length (distance or time).
4. Desire Line Diagram	Lines are drawn from zone centroids showing volume and direction of trips, used to identify major travel corridors.
5. Peak Hour Trip Analysis	Helps in determining the maximum load on roads, which aids in capacity planning and design.
6. Modal Split Analysis	Breaks down travel by transport mode (e.g., 40% bus, 30% private car, 20% two-wheeler), essential for transit planning.
7. Traffic Assignment	Estimated trips are assigned to specific routes based on shortest time, distance, or least cost—used for modeling.
8. Inference and Decision Making	Based on the analysis, planners can recommend improvements like new roads, widening, public transport enhancements, etc.





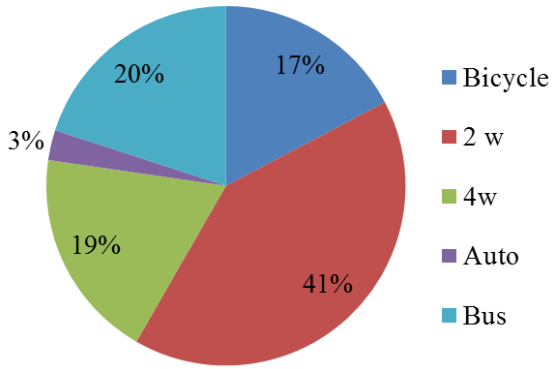




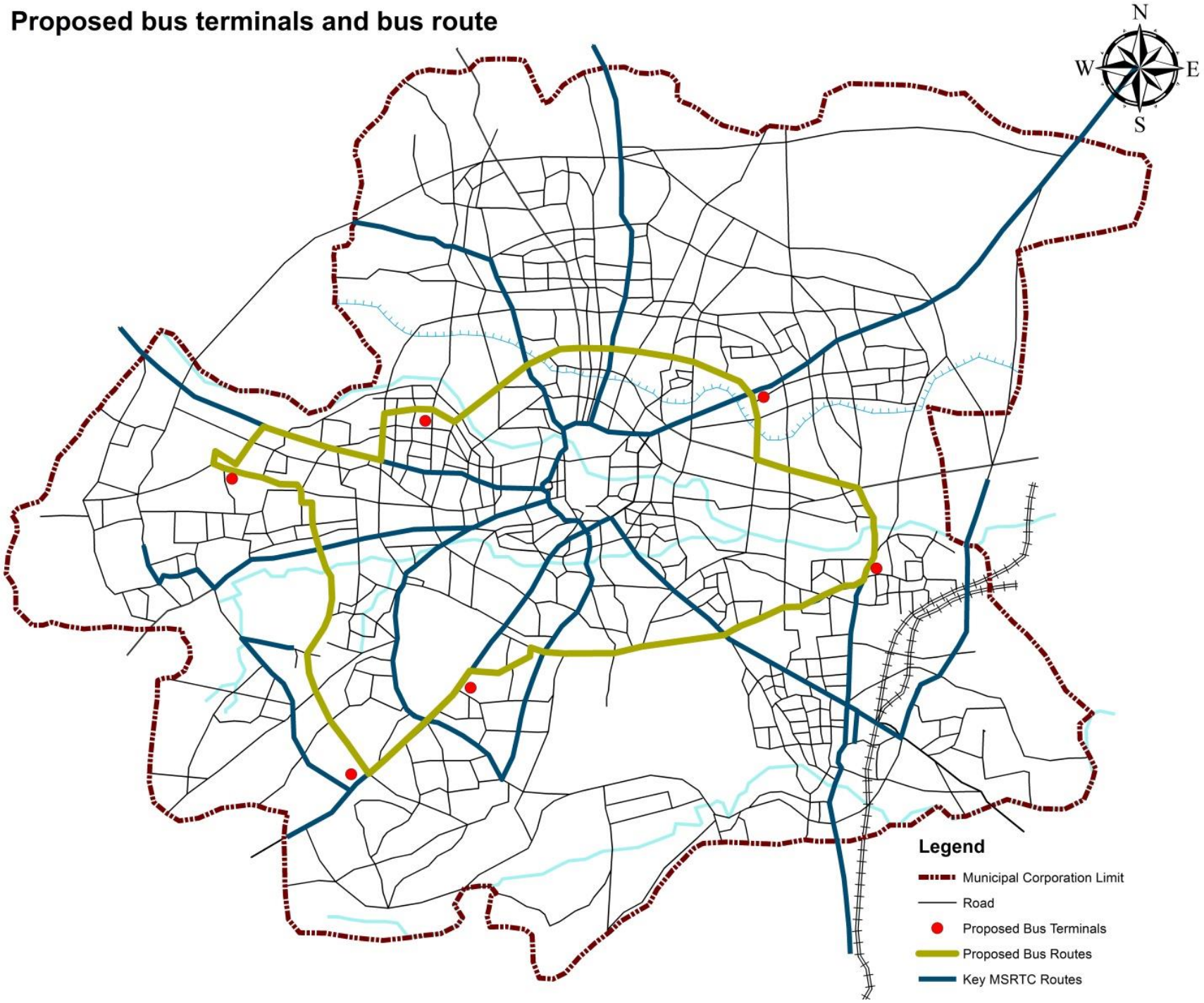


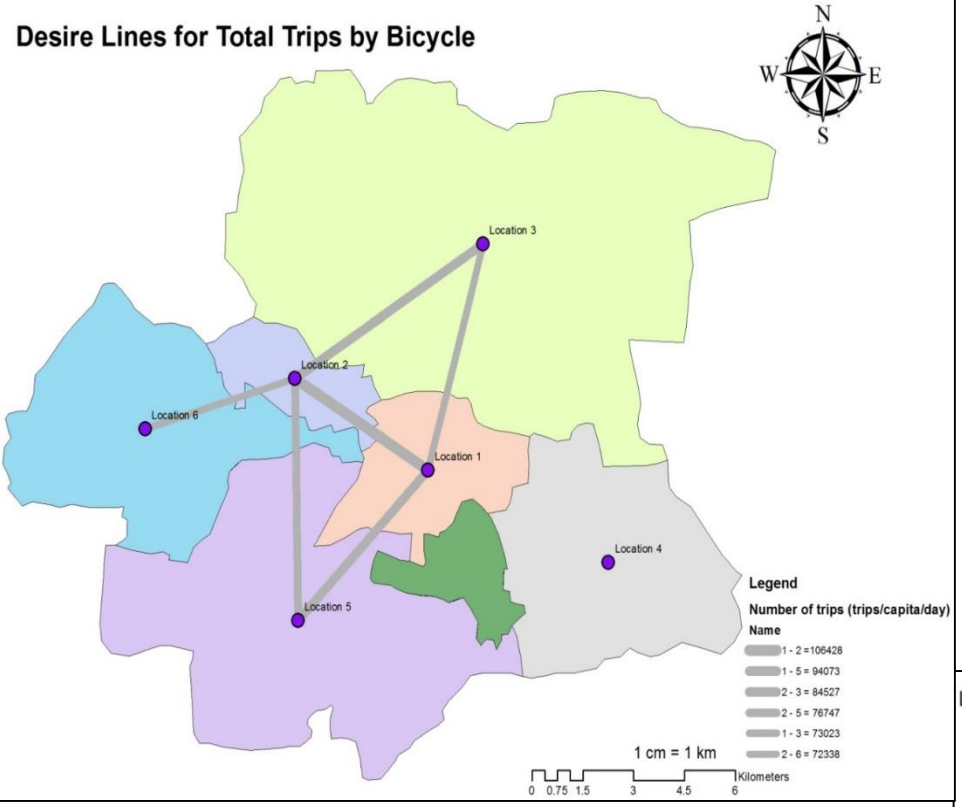
Probability of mode to be selected from zone 1 to other zones

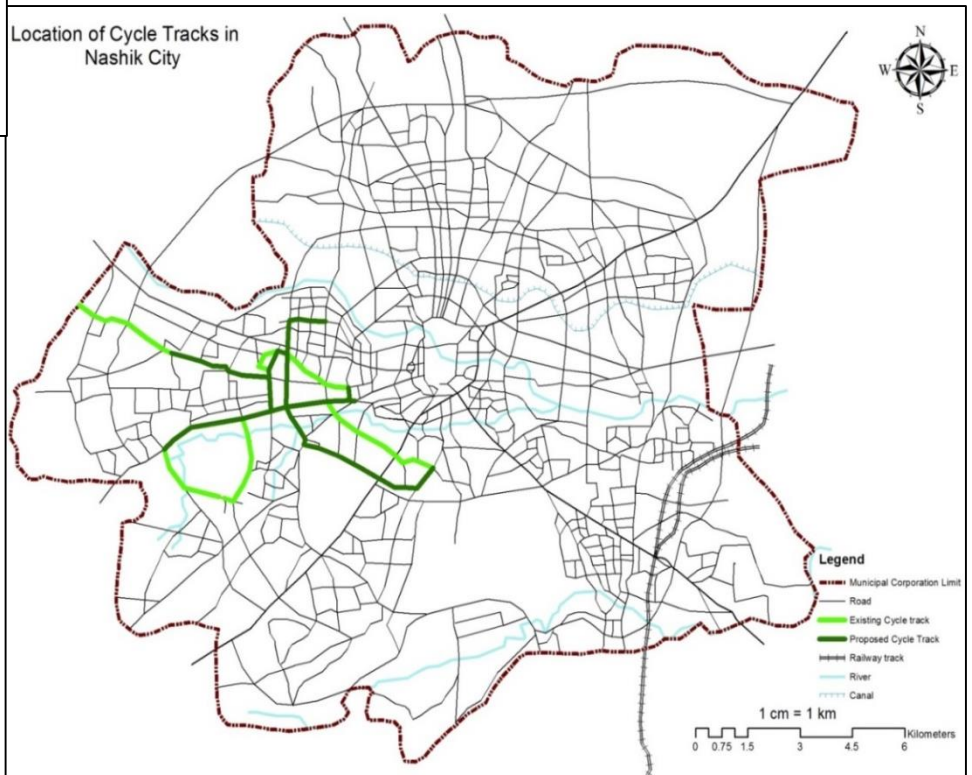
Modal Split For Worktrips



Proposed bus terminals and bus route




Desire Lines for Total Trips by Bicycle
N
W
E
S
Location 3
Location 2
Location 6
Location 1
Location 4
Location 5
Legend
Number of trips (trips/capita/day)
Name
1 - 2 = 106428
1 - 5 = 94073
2 - 3 = 84527
2 - 5 = 76747
1 - 3 = 73023
2 - 6 = 72338
1 cm = 1 km
0 0.75 1.5 3 4.5 6
Kilometers


Location of Cycle Tracks in
Nashik City
N
W
E
S
Legend
Municipal Corporation Limit
Road
Existing Cycle track
Proposed Cycle Track
Railway track
River
Canal
1 cm = 1 km
0 0.75 1.5 3 4.5 6
Kilometers

Flow

There are practically **two ways of counting the number of vehicles on a road.**

One is flow or volume, which is defined as the number of vehicles that pass a point on a highway or a given lane or direction of a highway during a specific time interval.

The measurement is carried out by counting the number of vehicles, nt, passing a particular point in one lane in a defined period t.

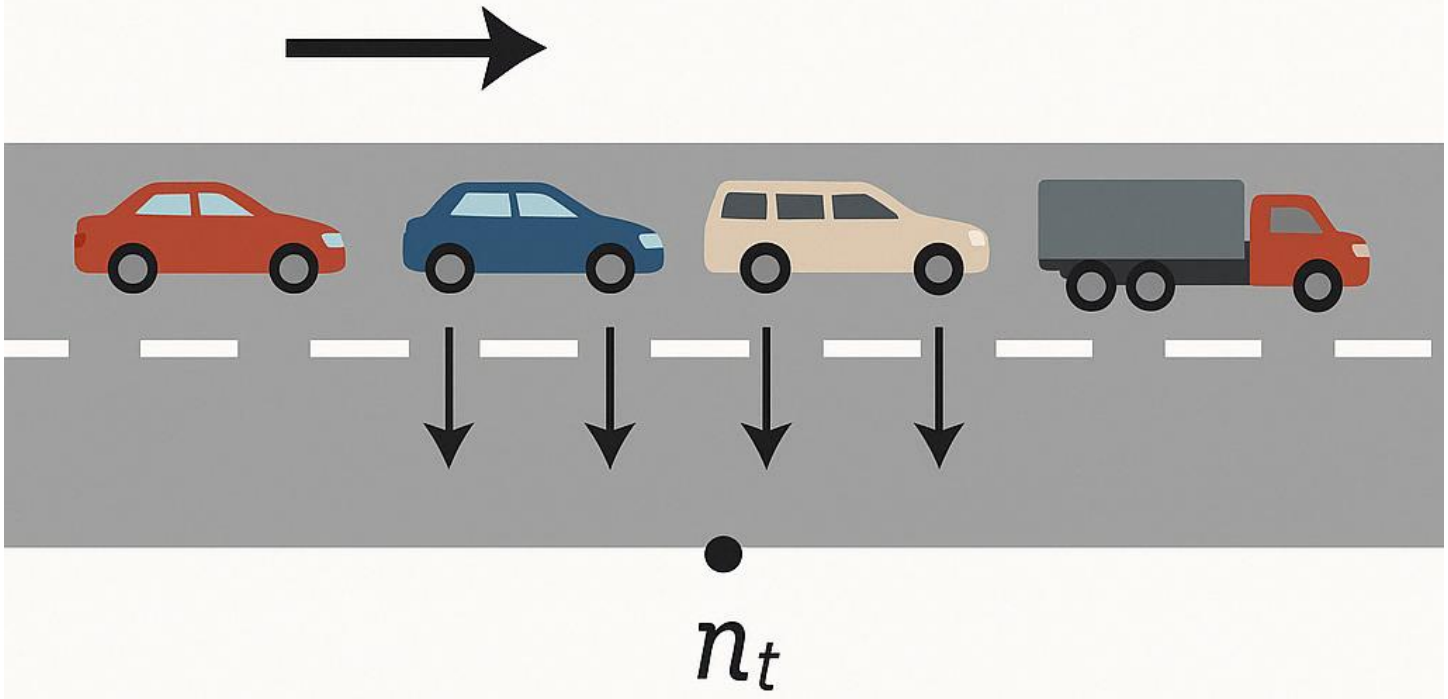
Then the flow q expressed in **vehicles/hour** is given by

$q = nt / t$

Flow is expressed in planning and design field taking a day as the measurement of time.

FLOW (OR VOLUME)

number of vehicles that pass a point on a highway or a given lane or direction of a highway during a specific time interval



Variations of Volume

- The variation of volume with time, i.e. month to month, day to day, hour to hour and within a hour is also as important as volume calculation. Volume variations can also be observed from season to season.
- Volume will be above average in a pleasant motoring month of summer, but will be more pronounced in rural than in urban area. But this is the most consistent of all the variations and affects the traffic stream characteristics the least.
- Weekdays, Saturdays and Sundays will also face difference in pattern. But comparing day with day, patterns for routes of a similar nature often show a marked similarity, which is useful in enabling predictions to be made.
- The most significant variation is from hour to hour. The peak hour observed during mornings and evenings of weekdays, which is usually 8 to 10 per cent of total daily ow or 2 to 3 times the average hourly volume.
- These trips are mainly the work trips, which are relatively stable with time and more or less constant from day to day.

Types of volume measurements

Term	Definition	Formula / Notes
AADT (Average Annual Daily Traffic)	The average 24-hour traffic volume at a location over a full 365-day year.	AADT = Total yearly traffic ÷ 365
AAWT (Average Annual Weekday Traffic)	The average 24-hour traffic volume on weekdays over a full year.	AAWT = Total weekday traffic in a year ÷ 260
ADT (Average Daily Traffic)	An average 24-hour traffic volume over a time period less than a year (e.g., week, month, season).	Valid only for the specific period it was measured
AWT (Average Weekday Traffic)	An average 24-hour traffic volume on weekdays for a time period less than a year (e.g., a month or season).	Valid only for the specific weekdays within the selected period

Design Hourly Traffic Volume

- Hourly traffic volume means the **volume of traffic of a particular hour**. So for a day, there will be 24 hourly traffic volumes. And for a year there will be 365×24 hourly volumes.

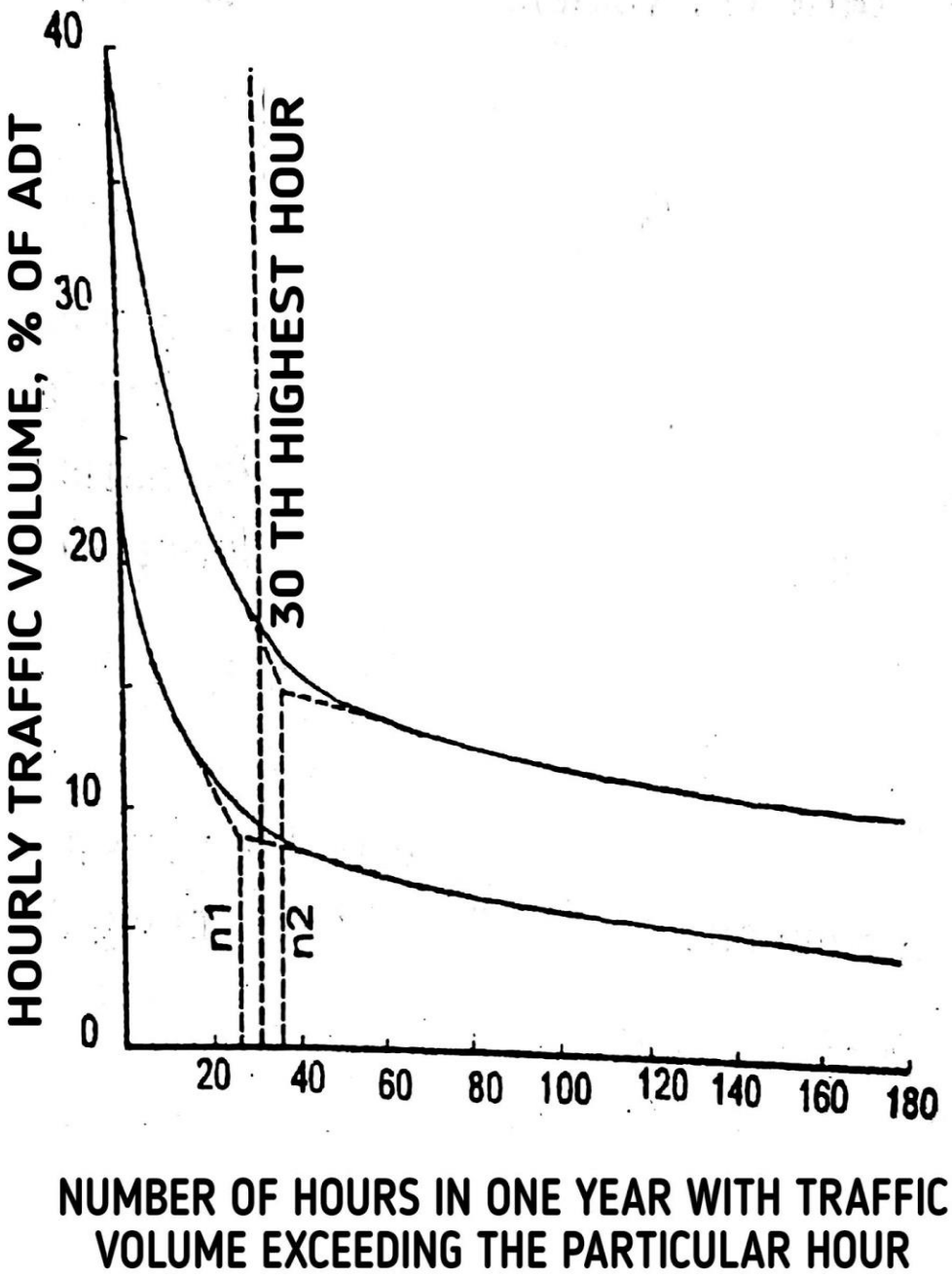
- The hourly volume **which is considered for design purpose** is called design hourly traffic volume. This value should be such that traffic volume at most of the time will be less than this.

- The maximum hourly volume or the peak hourly volume (i.e. the highest of all hourly volumes) of the year should be the obvious choice for this. But it is found to be too high and hence unnecessary to be considered for design as it's not an economical choice.

- The annual average hourly volume (AAHV) that is the average of all the hourly volumes of a year is not sufficient for the design purpose as there will be a considerable amount of time at which the traffic volume will be more than this value. **If the road will be designed for AAHV there will be severe congestion during a considerable period of the year.**

- So, the design hourly traffic volume should be between these two values and should be satisfactory from the consideration of the facility as well as the cost. "Thirtieth highest hourly traffic volume" is found to be satisfactory **while considering both facility and economic point of view.**

- Only 29 times in a year traffic volume will exceed this value that means there will be congestion only during 29 hours in the year and which is considered reasonable.
- The point to note that the cost will be much lesser when compared to the peak hourly volume. Thus the **30th highest hourly volume is generally adopted as the design hourly volume for the purpose of roadway facility design.**



Need for Traffic Volume Count Survey

Traffic Volume Survey is an essential part of Town Planning, especially for a town planner. It includes counting the number of vehicles passing through a survey station. The study of Classified Traffic Volume Count is to understand factors that form the basis of:

- Checking the efficiency/saturation of the road network by comparing current traffic volume with the calculated capacity or by identifying the level of service
- Establishing the use of the road network by vehicles of different categories, traffic distribution, PCU/vehicle value
- Need of median shifting or road widening

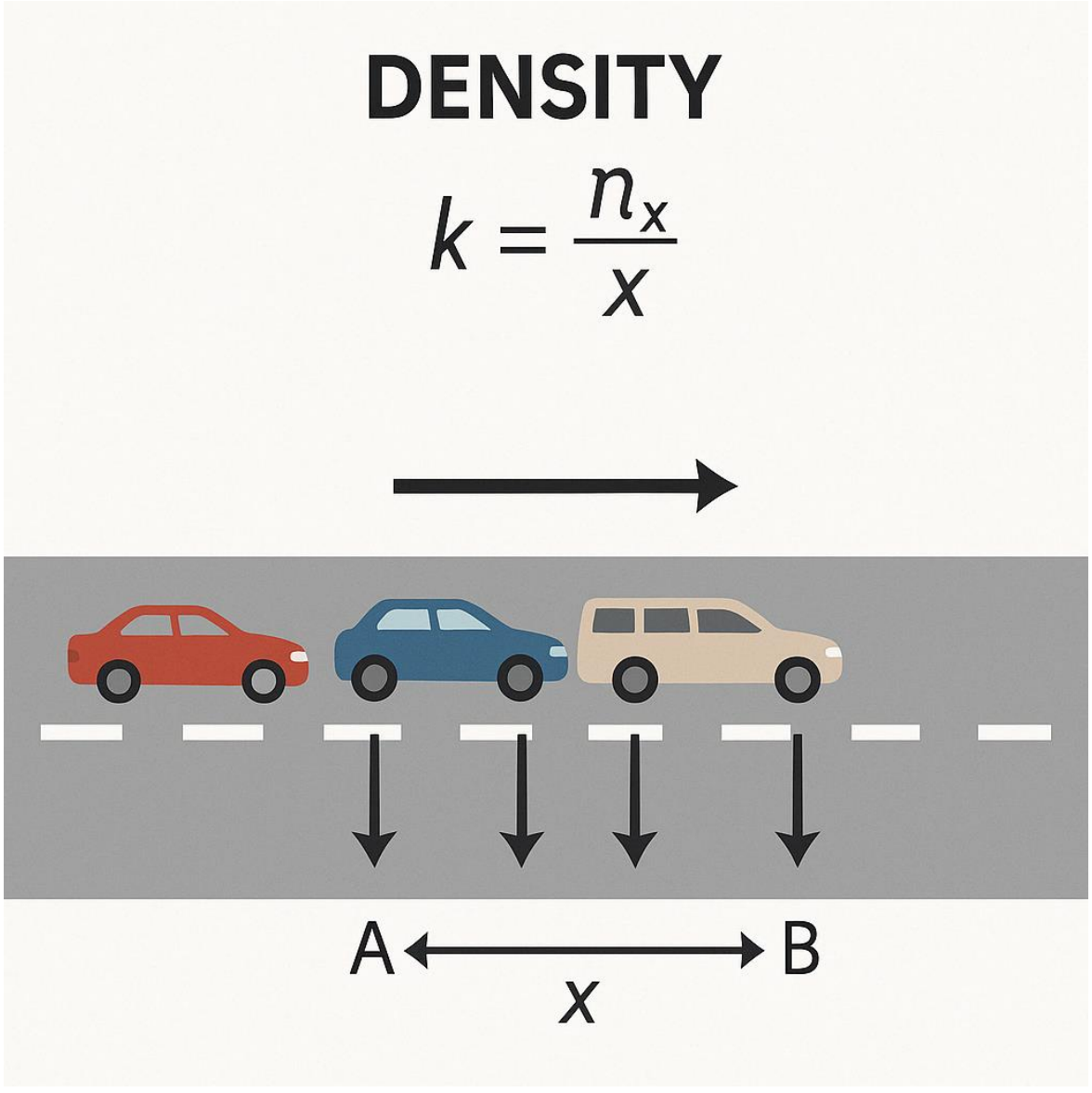
Purpose of Traffic Volume Count

The purpose of the classified traffic volume count is to draw inferences based on the data collected, to provide possible solutions and improvement suggestions for the problem identified.

1. Increase the efficiency and life of roads
2. Reduces traffic volume at a particular section
3. Provide better means for the development of infrastructures
4. Provide better means to utilize other roads in case of special events in the city
5. Provide an estimate of no vehicles against no of persons

Density

Density is defined as the number of vehicles occupying a given length of highway or lane and is generally expressed as **vehicles per km**.



Derived characteristics

Time headway (headway)

- The microscopic character related to volume is the time headway or simply headway.
- Time headway is defined as the **time difference between any two successive vehicles when they cross a given point.**
- Practically, it involves the measurement of time between the passage of one rear bumper and the next past a given point.

If all headways h in time period, t, over which flow has been measured are added then,

$$\sum_{1}^{n_t} h_i = t$$

But the flow is defined as the number of vehicles n_t measured in time interval t, that is,

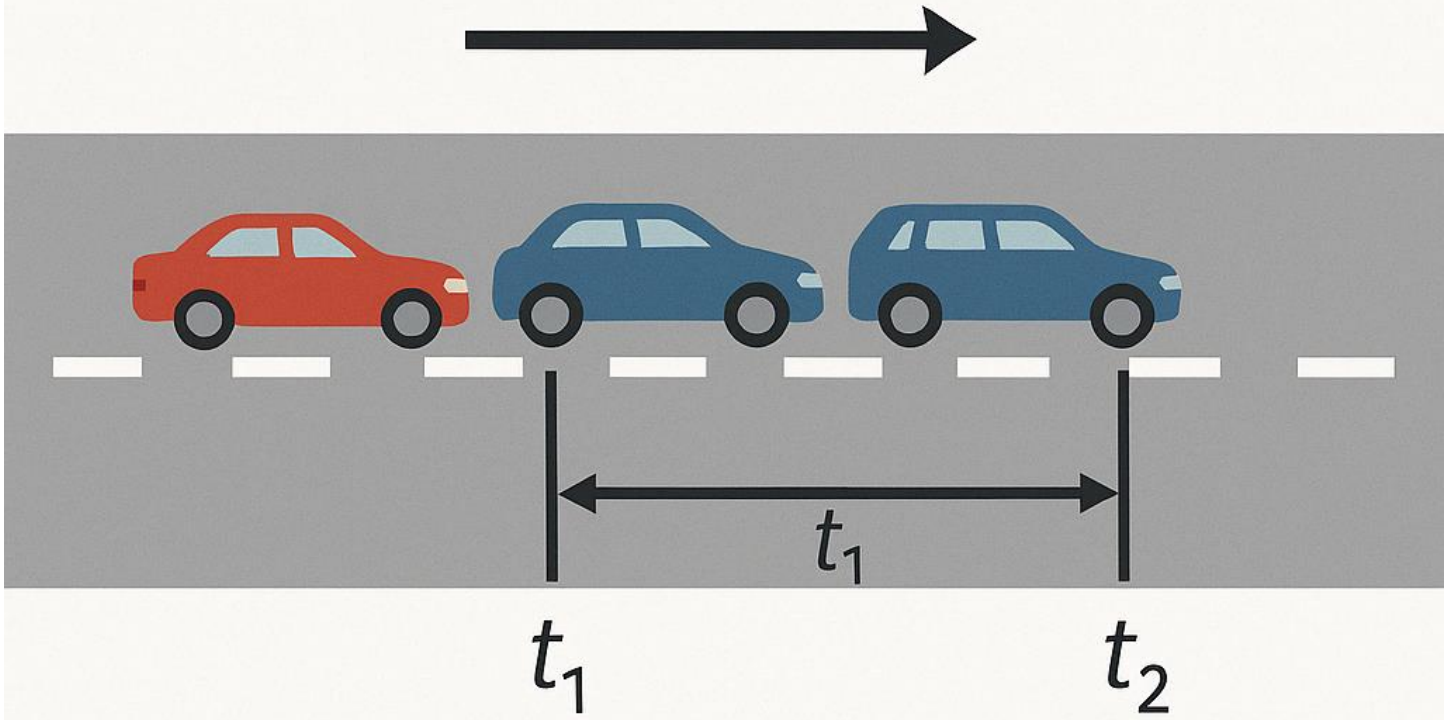
$$q = \frac{n_t}{t} = \frac{n_t}{\sum_1^{n_t} h_i} = \frac{1}{h_{av}}$$

where, h_{av} is the average headway. Thus **average headway is the inverse of flow.**

Time headway is often referred to as simply the headway

TIME HEADWAY

$$h = t_2 - t_1$$



Distance headway(Spacing)

- Another related parameter is the distance headway. It is defined as the distance between corresponding points of two successive vehicles at any given time.
- It involves the measurement from a photograph, the distance from rear bumper of lead vehicle to rear bumper of following vehicle at a point of time.
- If all the space headways in distance x over which the density has been measured are added,

$$\sum_{1}^{n_x} s_i = x$$

But the density (k) is the number of vehicles n_x at a distance of x, that is

$$k = \frac{n_x}{x} = \frac{n_x}{\sum_1^{n_x} s_i} = \frac{1}{s_{av}}$$

Where, sav is average distance headway. **The average distance headway is the inverse of density and is sometimes called as spacing.**

Travel time

- Travel time is defined as the time taken to complete a journey. As the speed increases, travel time required to reach the destination also decreases and vice-versa.
- Thus travel time is inversely proportional to the speed.
- However, in practice, the speed of a vehicle fluctuates over time and the travel time represents an average measure.

Time-space diagram

Time space diagram is **a convenient tool in understanding the movement of vehicles.** It shows the trajectory of vehicles in the form of a two dimensional plot. Time space diagram can be plotted for a single vehicle as well as multiple vehicles.

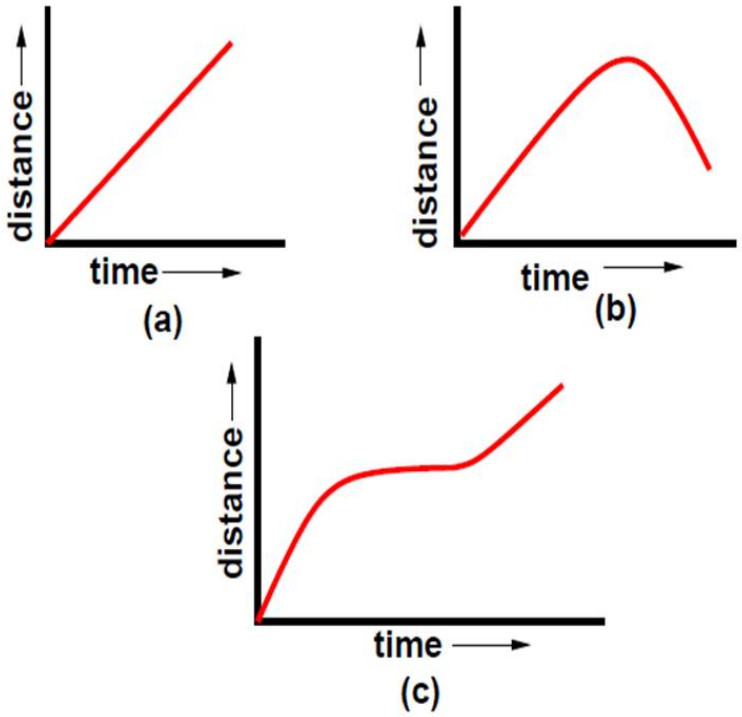


Figure 30:2: Time space diagram for a single vehicle

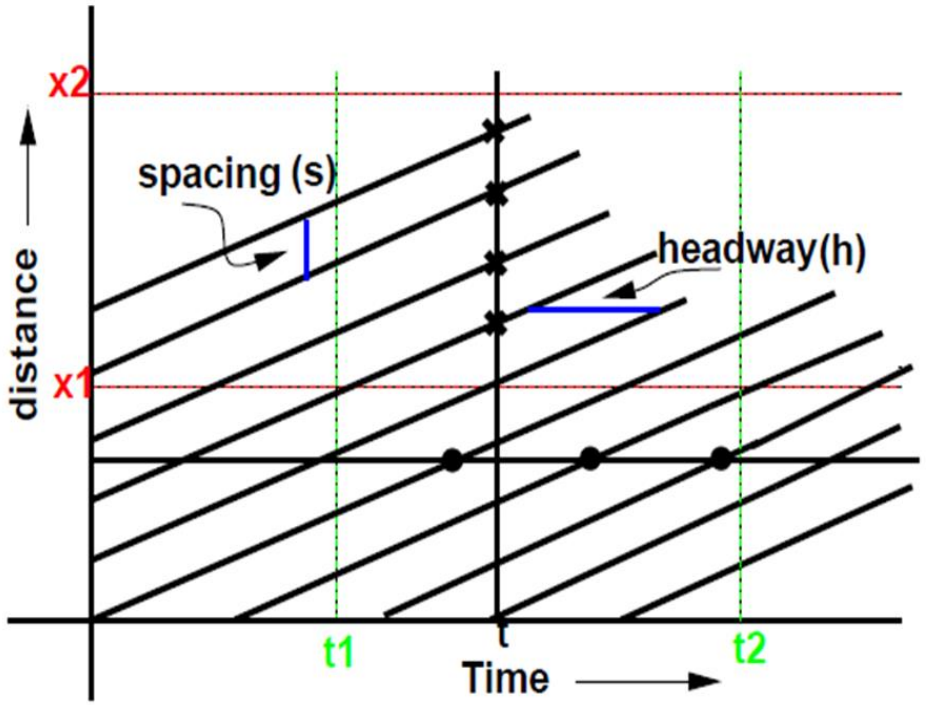


Figure 30:3: Time space diagram for many vehicles

Fundamental diagrams of traffic flow

The relation between flow and density, density and speed, speed and flow, can be represented with the help of some curves. They are referred to as the fundamental diagrams of traffic flow.

Flow-density curve

Some characteristics of an ideal flow-density relationship is listed below:

1. When the density is zero, flow will also be zero, since there is no vehicles on the road.
2. When the number of vehicles gradually increases the density as well as flow increases.
3. **When more and more vehicles are added, it reaches a situation where vehicles can't move. This is referred to as the jam density or the maximum density.** At jam density, flow will be zero because the vehicles are not moving.
4. There will be some density between zero density and jam density, when the flow is maximum

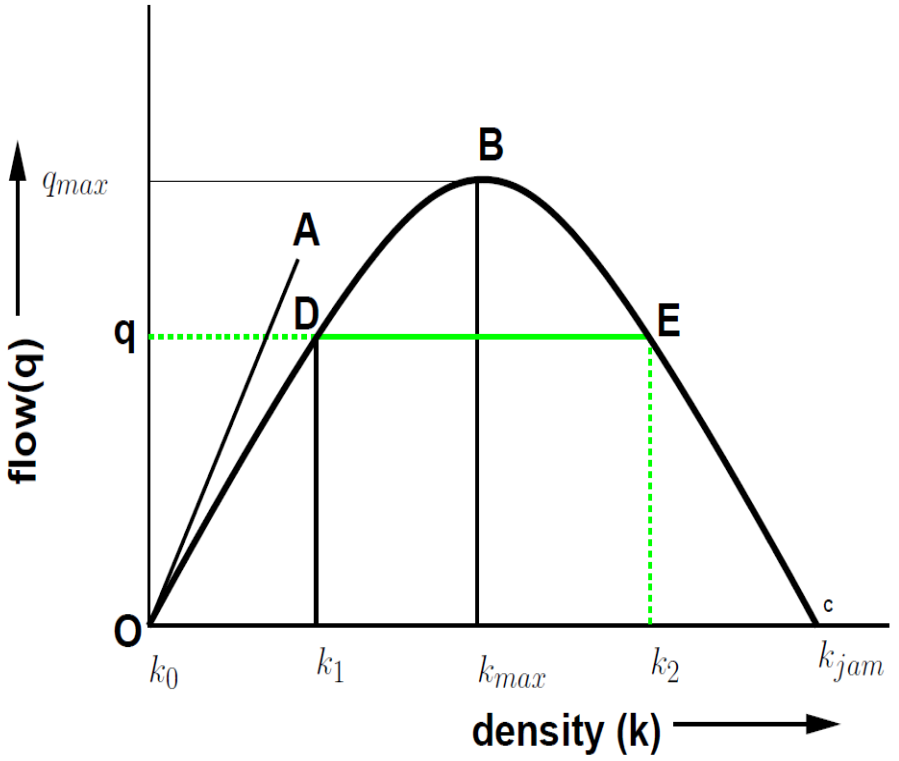


Figure 31:3: Flow density curve

Speed-density diagram

- Similar to the flow-density relationship, speed will be maximum, referred to as the free flow speed, and when the density is maximum, the speed will be zero.
- The most simple assumption is that this **variation of speed with density is linear** as shown by the solid line in figure 31:4.
- Corresponding to the zero density, vehicles will be showing with their desire speed, or free flow speed.
- When the density is jam density, the speed of the vehicles becomes zero.
- It is also possible to have non-linear relationships as shown by the dotted lines.

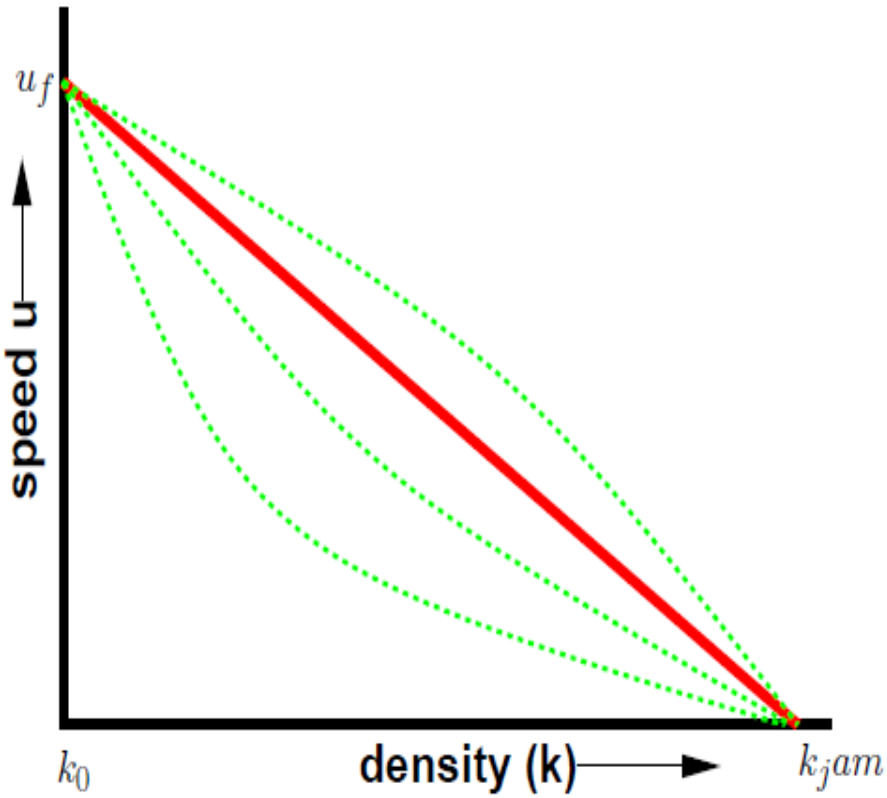


Figure 31:4: Speed-density diagram

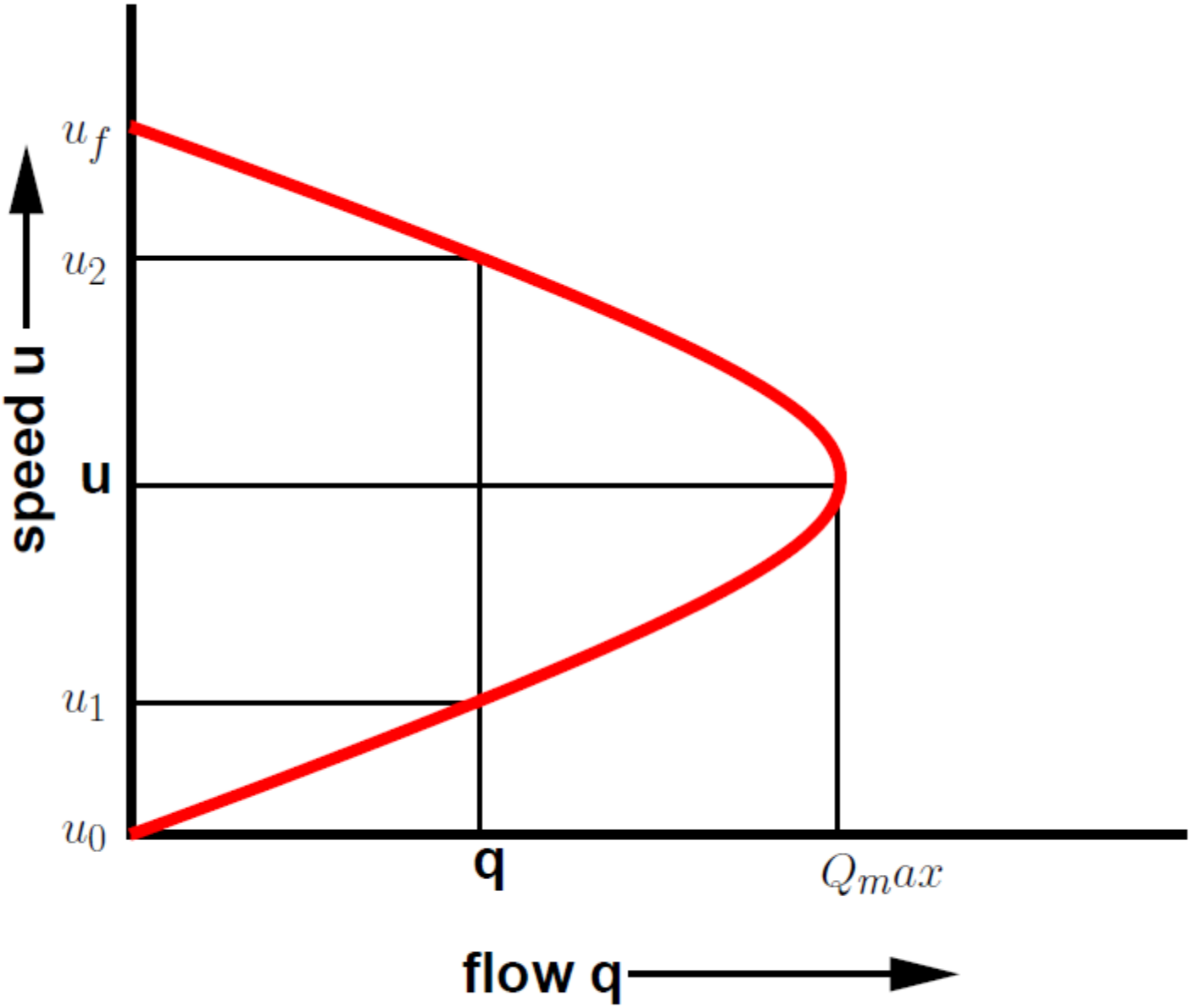


Figure 31:5: Speed-flow diagram

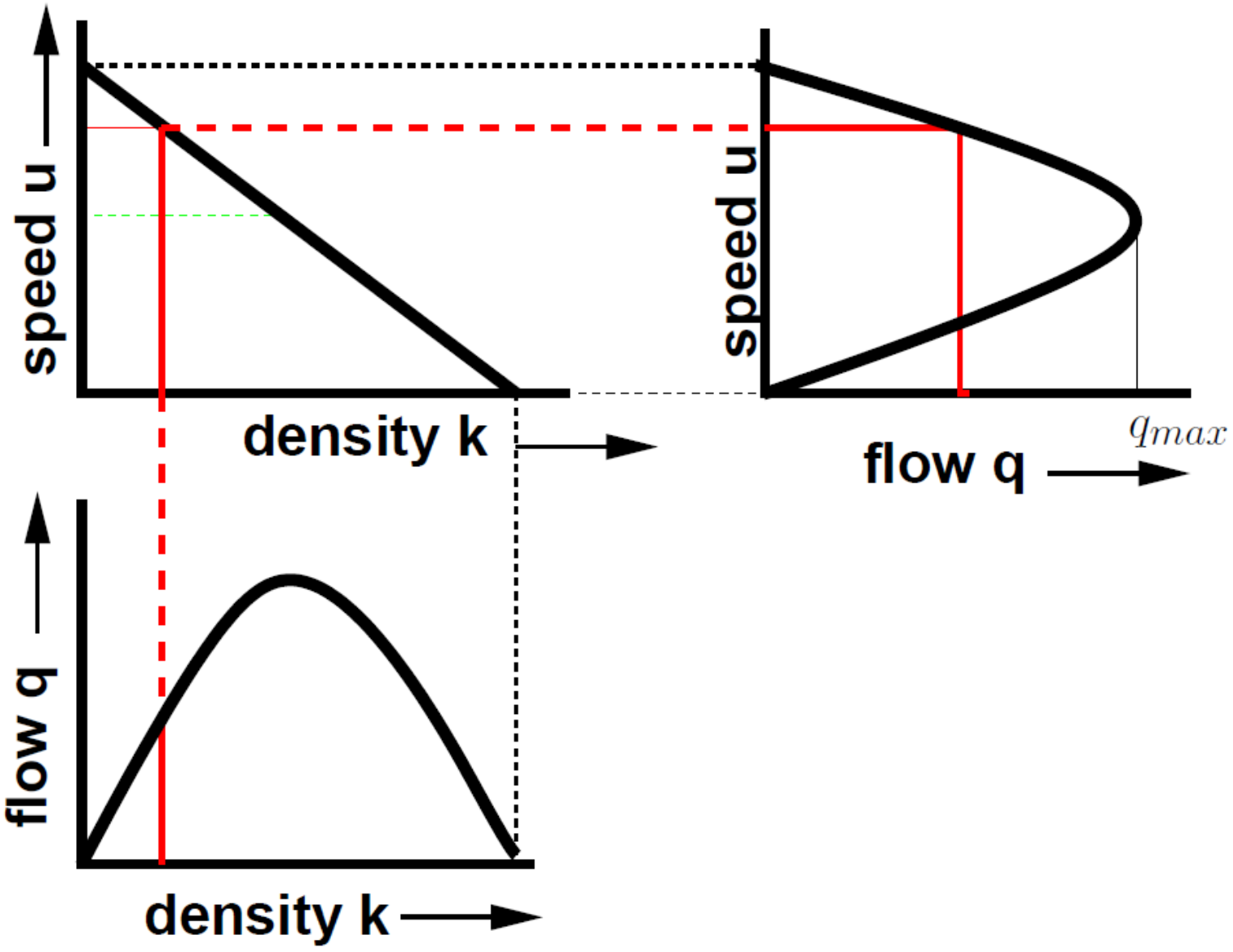


Figure 31:6: Fundamental diagram of traffic flow

Traffic stream models

To figure out the exact relationship between the traffic parameters, a great deal of research has been done over the past several decades. The results of these researches yielded many mathematical models.

Greenshield's macroscopic stream model

- Macroscopic stream models **represent how the behavior of one parameter of traffic flow changes with respect to another.**
- Most important among them is the relation between speed and density.
- The first and most simple relation between them is proposed by Greenshield.
- Greenshield assumed a linear speed-density relationship as illustrated in figure 33:1 to derive the model.
- The equation for this relationship is shown below

$$v = v_f - \left[\frac{v_f}{k_j}\right].k$$

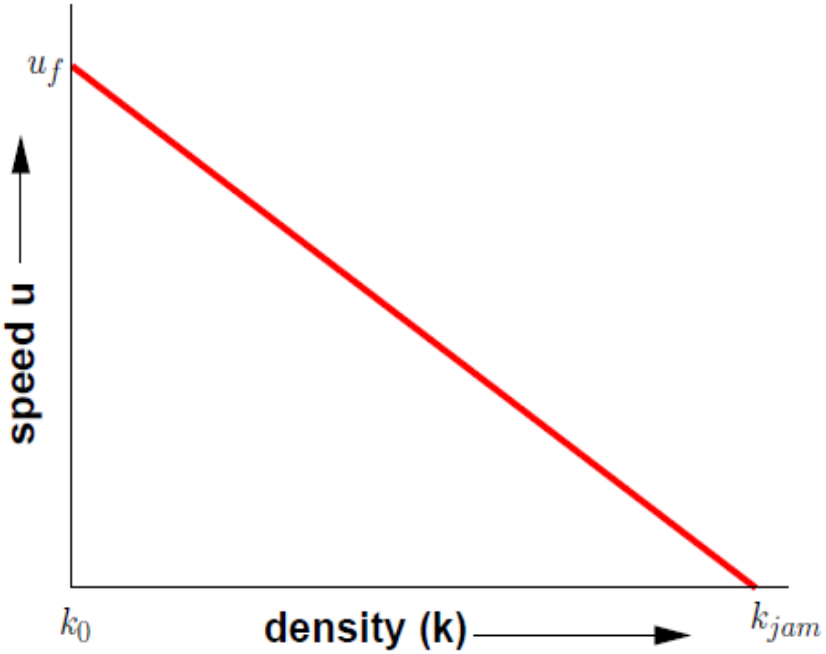


Figure 33:1: Relation between speed and density

where v is the mean speed at density k, vf is the free speed and kj is the jam density. This equation (33.1) is often referred to as the Greenshields' model. It indicates that when density becomes zero, speed approaches free flow speed (ie. v ! vf when k ! 0).

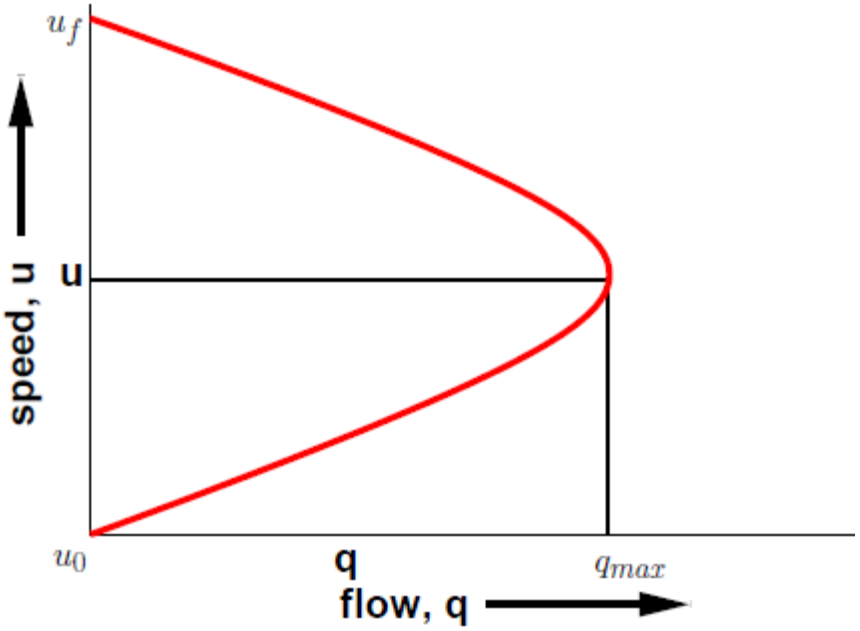


Figure 33:2: Relation between speed and flow

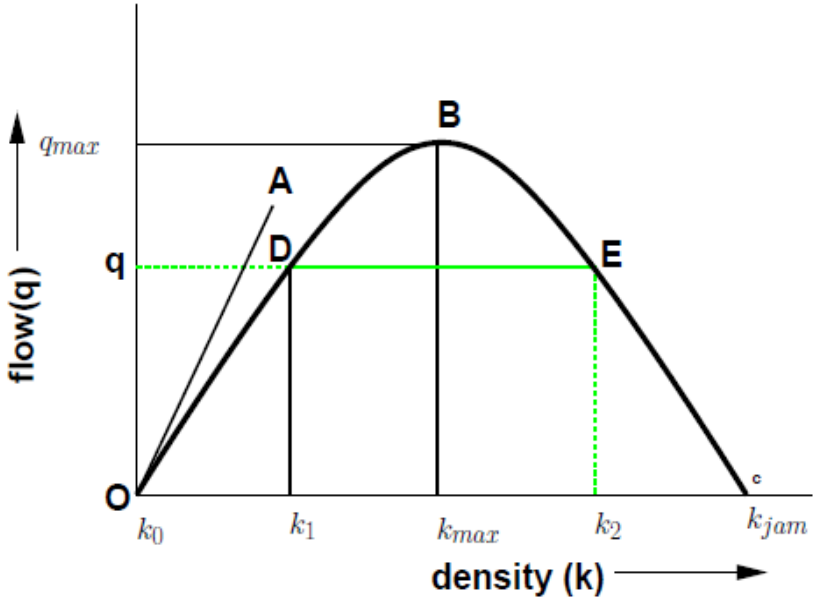


Figure 33:3: Relation between flow and density

Once the relation between speed and flow is established, the relation with flow can be derived.

This relation between flow and density is parabolic in shape and is shown in figure 33:3.

Also, we know that q = k * v

Denoting the density corresponding to maximum flow as ko,

$$k_0 = \frac{k_j}{2}$$

Therefore, density corresponding to maximum flow is half the jam density Once we get k_0, we can derive for maximum flow, q_{max}. Substituting equation 33.5 in equation 33.3

$$\begin{aligned} q_{max} &= v_f.\frac{k_j}{2} - \frac{v_f}{k_j}.\left[\frac{k_j}{2}\right]^2 \\ &= v_f.\frac{k_j}{2} - v_f.\frac{k_j}{4} \\ &= \frac{v_f.k_j}{4} \end{aligned}$$

Thus the maximum flow is one fourth the product of free flow and jam density. Finally to get the speed at maximum flow, v_0, substitute equation 33.5 in equation 33.1 and solving we get,

$$\begin{aligned} v_0 &= v_f - \frac{v_f}{k_j}.\frac{k_j}{2} \\ v_0 &= \frac{v_f}{2} \end{aligned} \tag{33.6}$$

Therefore, speed at maximum flow is half of the free speed.

Other macroscopic stream models

- In Greenshield's model, linear relationship between speed and density was assumed. **But in field we can hardly find such a relationship between speed and density.**
- Therefore, the validity of Greenshields' model was questioned and many other models came up. Prominent among them are Greenberg's logarithmic model, Underwood's exponential model, Pipe's generalized model, and multi regime models.

Greenberg's logarithmic model

Greenberg assumed a logarithmic relation between speed and density. He proposed,

$$v = v_0 \ln \frac{k_j}{k} \tag{33.11}$$

This model has gained very good popularity because this model can be derived analytically. (This derivation is beyond the scope of this notes). However, main drawbacks of this model is that as density tends to zero, speed tends to infinity. This shows the inability of the model to predict the speeds at lower densities.

Underwood's exponential model

Trying to overcome the limitation of Greenberg's model, Underwood put forward an exponential model as shown below.

$$v = v_f . e^{\frac{-k}{k_0}} \tag{33.12}$$

where v_fThe model can be graphically expressed as in figure 33:5. is the free flow speedm and k_o is the optimum density, i.e. the densty corresponding to the maximum flow. In this model, speed becomes zero only when density reaches infinity which is the drawback of this model. Hence this cannot be used for predicting speeds at high densities.

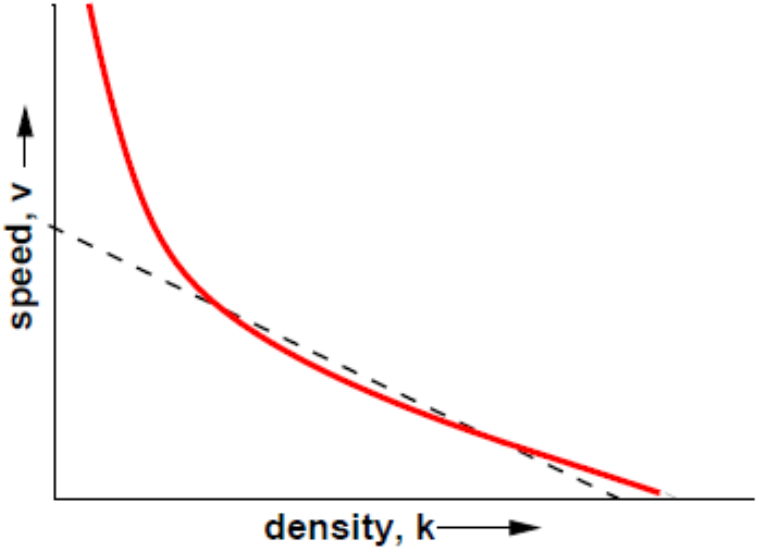


Figure 33:4: Greenberg's logarithmic model

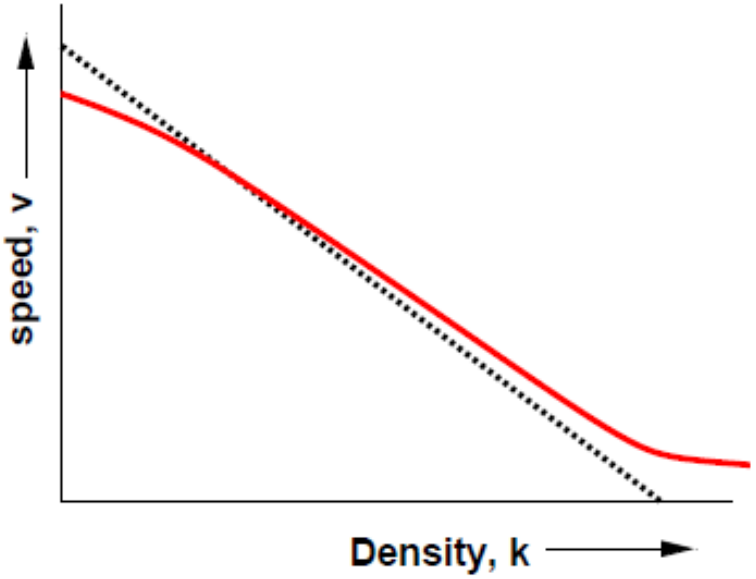


Figure 33:5: Underwood's exponential model

Passenger Car Unit (PCU)

S.No.	Component	Description
1	**Definition**	**PCU is a factor to convert different vehicle types into equivalent passenger cars for traffic flow analysis.**
2	**Purpose**	**Helps in analyzing capacity, design, and traffic performance in mixed traffic conditions.**
3	**Factors Affecting PCU**	
	a) Vehicle Dimensions	Larger vehicles → more road space → higher PCU
	b) Speed of Vehicle	Lower speed than cars → higher PCU
	c) Road Width	Narrow roads restrict overtaking → increases PCU
	d) Traffic Composition	More slow-moving vehicles → changes PCU of other vehicles
	e) Road Gradient / Curvature	Steep or curved roads reduce heavy vehicle speed → higher PCU
	f) Traffic Control	Signalized vs unsignalized intersections affect PCU
	g) Volume-to-Capacity Ratio	Higher congestion → PCU for large vehicles increases

For Rural Roads (Uncontrolled Intersections)

Vehicle Type	PCU Value
Passenger Car	1.0
Motor Cycle / Scooter	0.5
Auto Rickshaw	1.2
Light Commercial Vehicle	1.4
Truck / Bus	3.0
Tractor with Trailer	4.5
Cycle	0.5
Horse-drawn Vehicle	4.0

For Urban Roads (Midblock Sections)

Vehicle Type	PCU Value
Passenger Car	1.0
Motor Cycle / Scooter	0.5
Auto Rickshaw	1.0
Light Commercial Vehicle	1.5
Truck / Bus	2.5
Tractor with Trailer	4.0
Cycle	0.4
Animal-drawn Vehicle	6.0

Capacity

Definition	Capacity is the maximum number of vehicles that can pass a given point or section of a road during a specified time period (usually per hour) under prevailing roadway, traffic, and control conditions.
Units	Vehicles per hour or PCU per hour
Factors Affecting Capacity	- Road width - Traffic composition - Shoulder condition - Road geometry (gradient, curvature) - Access control - Surface condition - Environmental factors
Capacity Formula (Simplified)	Capacity = Basic Capacity × Adjustment Factors
Basic Capacity (Ideal Conditions)	Provided in IRC:106-1990 (values below)

IRC Recommended Basic Capacity (Ideal Conditions)

Road Type	Capacity (PCU/hr/direction)
Single-lane (unidirectional)	1,000 – 1,200
Two-lane (undivided)	2,000 – 2,400
Four-lane (undivided)	4,000 – 4,800
Four-lane (divided)	3,000 – 3,500 per direction
Six-lane (divided)	5,400 – 5,800 per direction
Urban arterial road	1,800 – 2,400 per direction

Level of Service (LOS)

Definition	LOS is a qualitative measure describing operational conditions within a traffic stream and the perception of drivers.
Criteria	Speed, travel time, freedom to maneuver, comfort, convenience, and traffic interruptions.
Range	LOS is classified from A to F

Levels of Service (LOS Classification)

LOS	Description	Volume/Capacity (V/C) Ratio
A	Free flow, low volume, high speed	0.00 – 0.35
B	Reasonably free flow	0.35 – 0.50
C	Stable flow, speed affected by other vehicles	0.50 – 0.75
D	Approaching unstable flow	0.75 – 0.90
E	Unstable flow, operating capacity reached	0.90 – 1.00
F	Forced flow, breakdown condition	> 1.00

Traffic Operations

Traffic engineering is a branch of civil engineering **that uses engineering techniques to achieve the safe and efficient movement of people and goods on roadways.**

- It focuses mainly on research for safe and efficient traffic flow, such as road geometry, sidewalks and crosswalks, cycling infrastructure, traffic signs, road surface markings and traffic lights.
- Traffic engineering deals with the functional part of transportation system, except the infrastructures provided.
- Traditionally, road improvements have consisted mainly of building additional infrastructure.
- However, dynamic elements are now being introduced into road traffic management.
- Dynamic elements have long been used in rail transport.
- These include sensors to measure traffic flows and automatic, interconnected, guidance systems to manage traffic

TRAFFIC REGULATION AND CONTROL

Definition	Traffic regulation and control refer to the rules, devices, and systems implemented to ensure safe, orderly, and efficient movement of people and goods on roads.
Objectives	
Ensure smooth and orderly movement of traffic on roads **Enhance safety** for all road users—drivers, passengers, cyclists, pedestrians **Minimize conflicts** between vehicles and between vehicles and pedestrians **Improve efficiency** of road network usage **Regulate speed and flow** of vehicles based on road geometry and environment **Reduce congestion and delays** especially at intersections and bottlenecks **Facilitate pedestrian movement** and provide safe crossing opportunities **Provide clear guidance** to road users through signs, markings, and signals **Assist enforcement** of traffic laws and rules	
Categories	**Support public transport** and emergency services movement efficiently

I. TRAFFIC REGULATION

a) Speed Regulations	- Speed limits as per road class, location - Enforced using signboards and speed cameras
b) Vehicle Regulations	- Licensing, registration, load limits, emission norms
c) Driver Regulations	- Licensing, minimum age, fitness, driving behavior
d) Parking Regulations	- No-parking zones, duration-based parking, off-street/on-street classification
e) One-Way Rules	- Used to reduce conflicts in congested areas
f) Overtaking & Lane Use	- Based on road width, visibility, markings
g) Load Restrictions	- Axle load limits, height and width clearance as per IRC norms

II. TRAFFIC CONTROL

- Signs	- Regulatory (mandatory/prohibitory) - Warning signs - Informatory signs (As per IRC:67)
- Signals	- Manual, fixed-time, traffic-actuated signals at intersections
- Road Markings	- Centre lines, stop lines, pedestrian crossings (As per IRC:35)
- Traffic Islands	- Channelizing, rotary, refuge islands to manage flow
- Barriers and Bollards	- Used to restrict or guide movement

Traffic Control Devices:

The improvement made & facilities provided to the road users so as to obtain efficient, free & rapid flow of traffic with less number of accidents are known as traffic control devices.

Types of Traffic Control Devices.

1) Traffic Signs or Road Signs
2) Traffic Markings or Road Markings
3) Traffic Islands
4) Traffic Signals

Objective:

- The main object of providing traffic control devices on road is **to provide safe, convenient & economical transportation** of persons & goods.

Classification of Road Signs:

1) **Regulatory or Mandatory Signs:** Gives information to road users certain laws, regulations & prohibitions.

2) **Warning or Cautionary Signs:** Used to warn the road users in advance of certain hazardous condition that exists on the road way.

3) **Informatory Signs :**Used to guide the road users along routes, inform them of destination, distance, etc.

Type	Usage	Symbol
Mandatory Signs	Signs that **give orders.** These are represented by red circles.	
Cautionary Signs	Signs that **give warnings.** These are represented by red triangles	
Informative Signs	Signs that **give information.** These are represented by blue rectangles.	

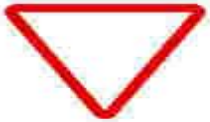
Regulatory Signs



STOP



NO ENTRY



GIVE WAY



ALL MOTOR VEHICLES PROHIBITED



NO MOTORCYCLES



NO BICYCLES



NO BULLOCK CARTS



NO HEAVY VEHICLES



ONE - WAY TRAFFIC



NO STRAIGHT AHEAD



NO VEHICLES IN BOTH DIRECTION



NO LEFT TURN



NO RIGHT TURN



NO OVERTAKING



NO U-TURN



HEIGH LIMIT



SPEED LIMIT



WEIGHT LIMIT



LENGHT LIMIT



NO PARKING



HORN PROHIBITED



NO PEDESTRIANS



LEFT CURVE



RIGHT CURVE



TRAFFIC SIGNALS AHEAD



LEVEL CROSSING WITH BARRIER AHEAD



UNEVEN ROAD



LEVEL CROSSING WITHOUT BARRIER AHEAD



NARROW BRIDGE



ROUNDABOUT AHEAD

Warning Signs



CW057



CW058



CW059



CW060



CW061



CW064



CW065



CW066



CW067



CW068



CW071



CW072



CW073



CW074



CW075



CW078



CW078



CW080



CW081



CW082



CW085



CW086



CW087



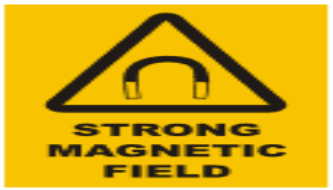
CW088



CW089



CW092



CW093



CW094



CW095



CW096

Informatory Signs






पलवल
Palwal
53 Km
अलीगढ़
Aligarh
74 Km
मथुरा
Mathura
100 Km
आगरा
Agra
159 Km


SPEED LIMIT
ENFORCED BY
AIRCRAFT




PHONE DOWN
IT'S THE LAW

Road Markings

Objective

1. To **guide and control traffic** safely and efficiently
2. To **channelize traffic movement** in desired paths
3. To **indicate regulations** (like no overtaking, lane discipline)
4. To assist in **safe pedestrian crossing**
5. To **enhance visibility** during night and adverse weather

Type	
Longitudinal Markings	Along the direction of travel. Used for lane division and overtaking control.
Transverse Markings	Across the direction of travel. Used for stop lines, pedestrian crossings, etc.
Hazard Markings	Warn drivers about obstructions, diversions, or merging lanes.
Block Markings	Symbols, arrows, and other informative icons on road surface.
Word Messages	Writings like “STOP”, “SLOW”, “SCHOOL” etc., for specific instructions.
Parking Markings	Define parking zones, no-parking areas, and bays.
Facility Markings	Markings for bus bays, cycle tracks, loading zones, etc.

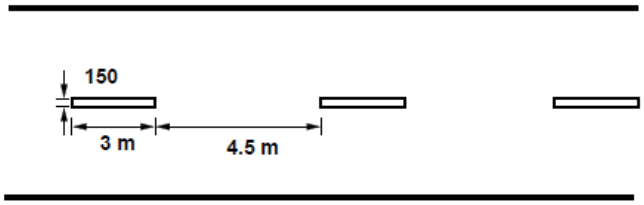


Figure 37:1: Centre line marking for a two lane road

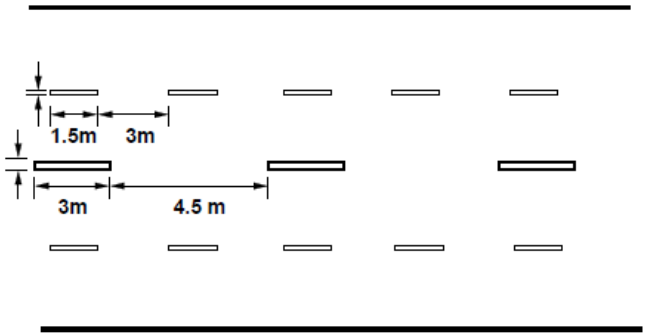


Figure 37:2: Centre line and lane marking for a four lane road

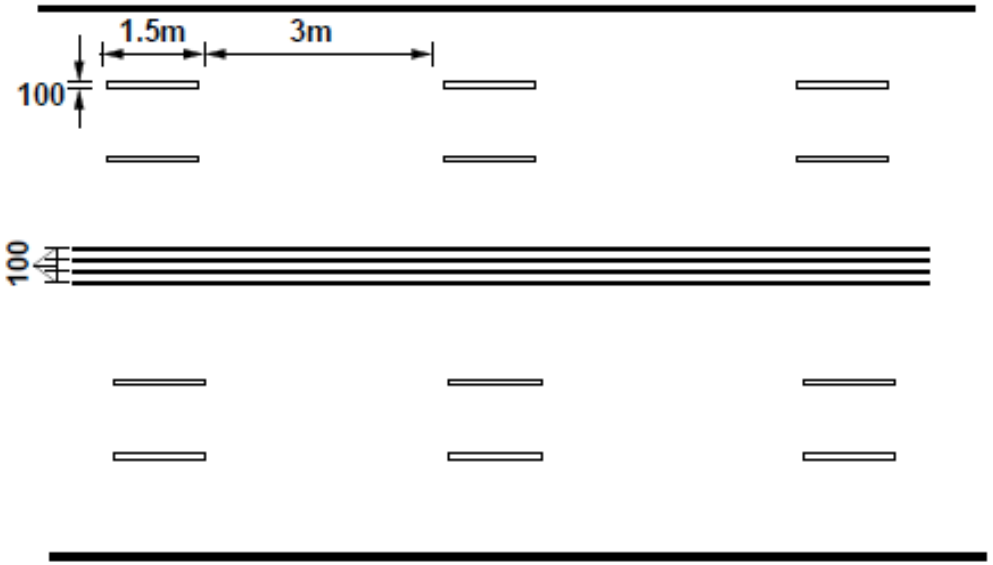


Figure 37:3: Double solid line for a two lane road

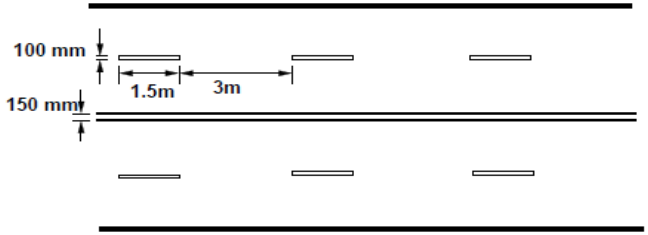


Figure 37:4: Centre barrier line marking for four lane road

Longitudinal Markings

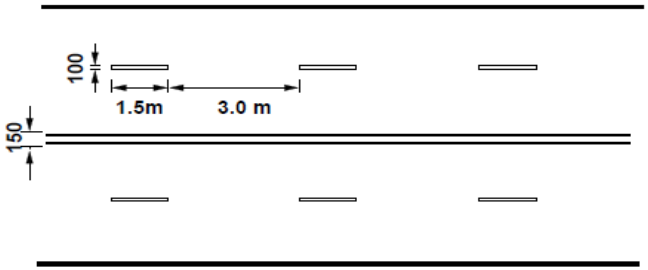


Figure 37:5: Lane marking for a four lane road with solid barrier line

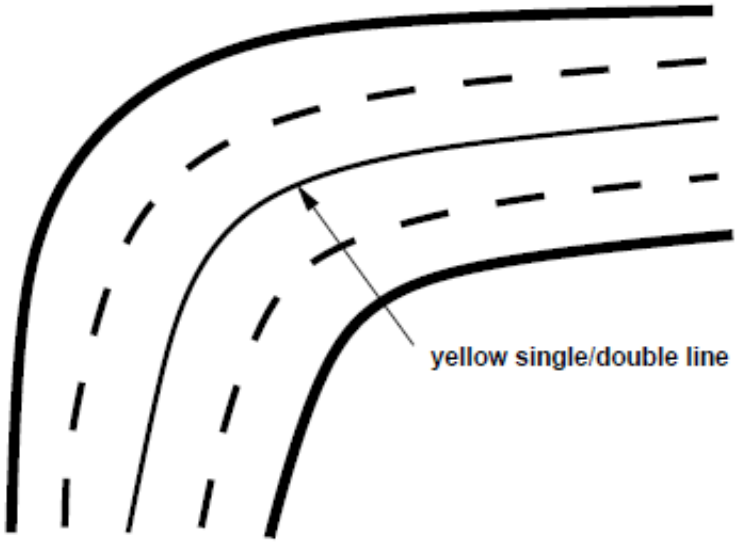


Figure 37:7: Barrier line marking for a four lane road

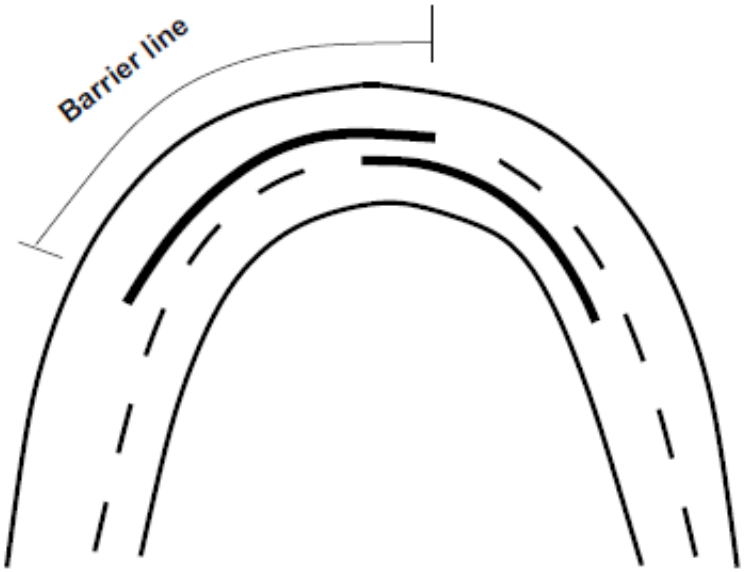


Figure 37:8: No passing zone marking at horizontal curves

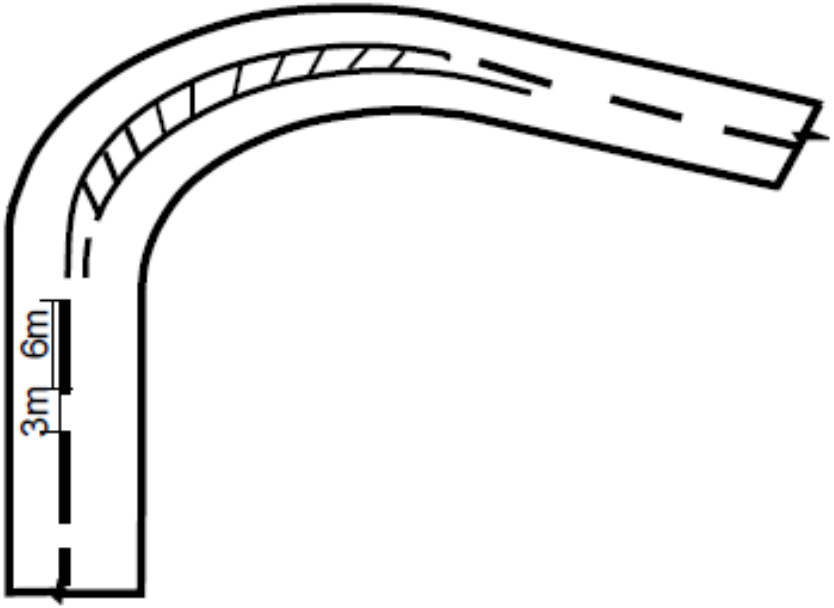


Figure 37:9: Warning line marking for a two lane road

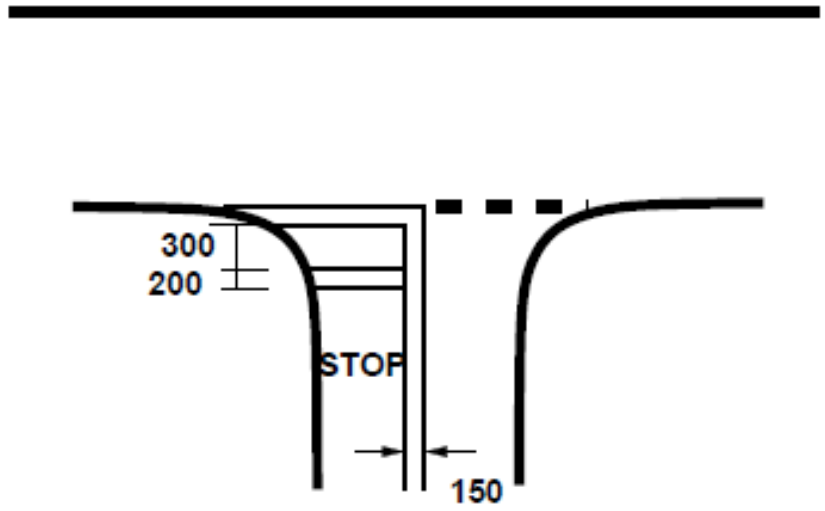


Figure 37:10: Stop line marking near an intersection

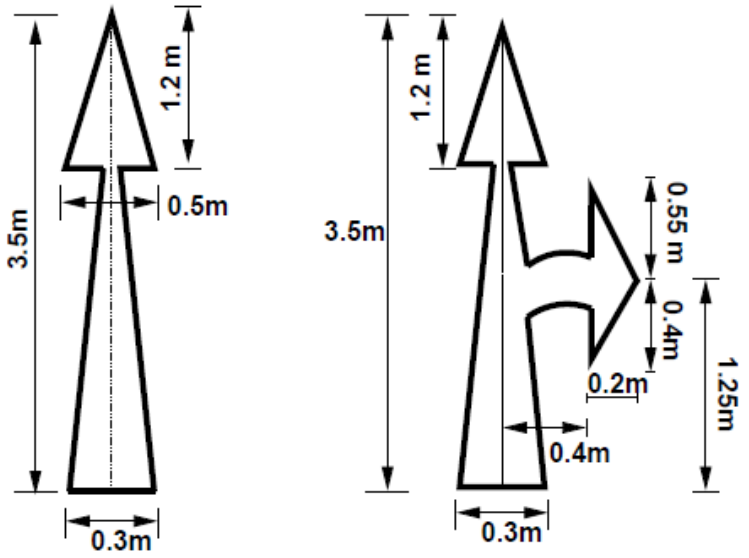


Figure 37:12: Directional arrow marking

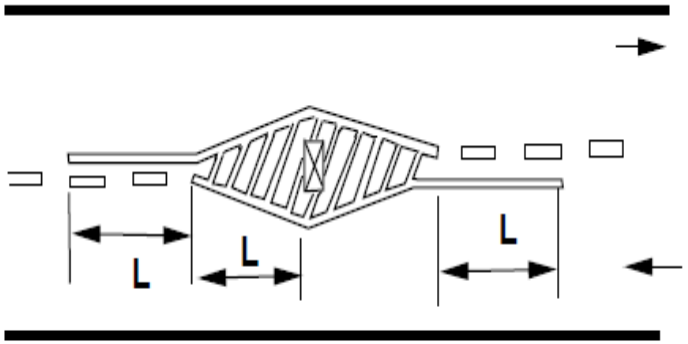
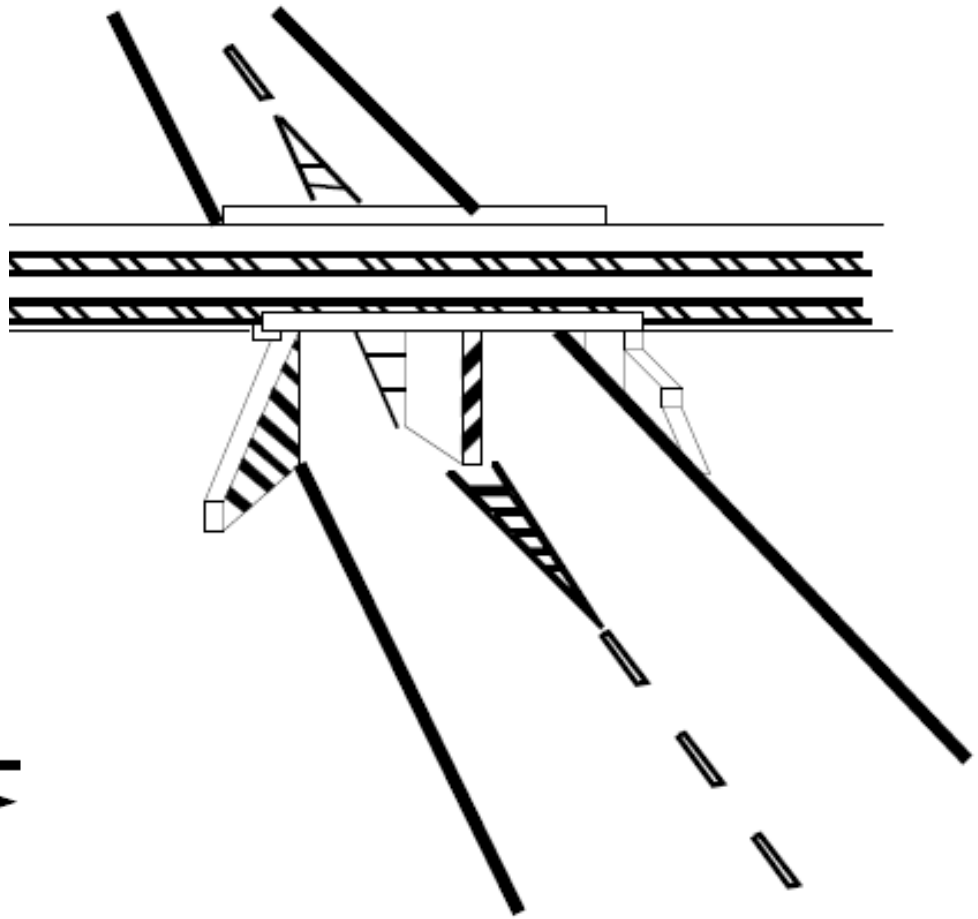


Figure 37:15: Approach marking for obstructions on the road way



Marking for objects adjacent to the road way

TRAFFIC ISLANDS

Objective

1. To **channelize traffic flow** in a systematic manner
2. To **separate conflicting traffic movements**
3. To provide **safe refuge areas** for pedestrians
4. To help in **reducing vehicle speed** at intersections
5. To **control entry and exit points** at intersections
6. To improve **intersection efficiency and safety**

✓ Advantages

1. Improved Safety
2. Better Guidance
3. Pedestrian Safety
4. Speed Control
5. Traffic Discipline
6. Aesthetic Value

✗ Disadvantages

1. Space Requirement
2. Construction Cost
3. Obstruction Risk
4. Drainage Issues
5. Driver Confusion

1. Divisional Island

Separates opposing traffic streams on undivided roads or at median openings



2. Channelizing Island

Directs traffic into specific paths (e.g., left-turn lanes, slip roads)



3. Rotary Island

Central island in a roundabout; facilitates circular movement around the intersection



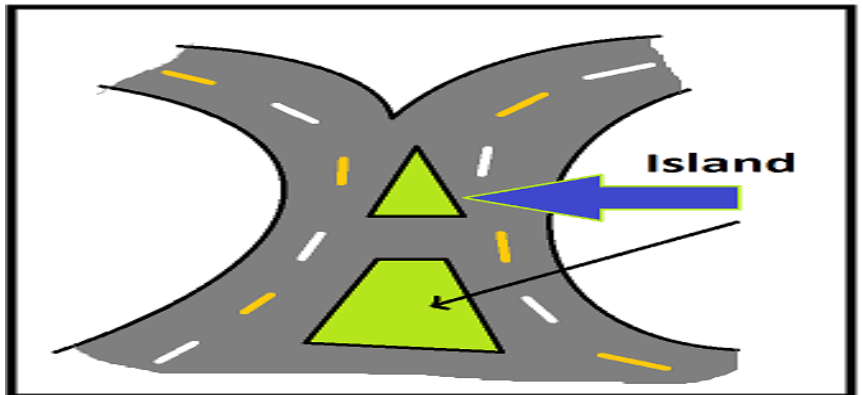
4. Pedestrian Refuge Island

Provides safe waiting zone for pedestrians while crossing wide roads



5. Directional Island

Used to guide or restrict movement in one or more directions, often triangular in shape



TRAFFIC SIGNALS

Objective

1. To **assign right-of-way** to different movements of traffic at intersections
2. To **reduce conflicts** at intersections and ensure safety
3. To **increase intersection capacity** through organized movement
4. To **regulate pedestrian movement** safely across intersections
5. To **minimize delays and accidents** at busy or complex intersections
6. To **facilitate orderly and efficient flow** of vehicles in urban networks

Advantage

1. Improves safety by **reducing collision points**
2. Allows **efficient use of intersection** during peak hours
3. Provides **safe crossing** for pedestrians
4. Helps in **systematic control of large traffic volumes**
5. Can be coordinated with other signals for **smooth flow** on urban arterials

Disadvantage

1. May cause **unnecessary delay** during low traffic periods
2. Can result in **rear-end collisions** if not timed correctly
3. **Non-compliance by drivers** can lead to confusion and risk
4. Installation and maintenance can be **costly**
5. Improperly designed signals may cause **congestion and driver frustration**

TYPES OF TRAFFIC SIGNALS

Type	
Fixed Time Signals	Signal timings are **pre-set and do not change with traffic volume**. Best for uniform traffic.
Manual Signals	Operated **by traffic police** during special or emergency conditions.
Traffic Actuated Signals	Adjust signal timing **based on traffic volume using detectors or sensors.**
Semi-Actuated Signals	**One or more approaches are equipped with detectors**; minor roads trigger green.
Fully Actuated Signals	**All approaches** are equipped **with detectors for dynamic timing**.
Pedestrian Signals	Signals specifically designed to **assist safe pedestrian crossing**.
Flashing Signals	**Used during off-peak hours or emergencies** (Yellow = caution, Red = stop).
Coordinated Signals	Signals along a corridor operate in sync for smooth progression of traffic (green waves).

Design of Signals

- Traffic signals are one of the most effective and flexible active control of traffic and is widely used in several cities world wide.
- The conflicts arising from movements of traffic in different directions is addressed by time sharing principle.
- The advantages of traffic signal includes an orderly movement of traffic, an increased capacity of the intersection and requires only simple geometric design.
- However, the disadvantages of the signalized intersection are large stopped delays, and complexity in the design and implementation.
- Although the overall delay may be lesser than a rotary for a high volume, a user may experience relatively high stopped delay.

Design of Signals

- A number of definitions and notations need to be understood in signal design. They are discussed below:
- **Cycle**: A signal cycle is one complete rotation through all of the indications provided.
- **Green interval**: It is the green indication for a particular movement or set of movements and is denoted by Gi.
  - This is the actual duration the green light of a traffic signal is turned on.
- **Cycle length**: Cycle length is the time in seconds that it takes a signal to complete one full cycle of indications.
  - It indicates the time interval between the starting off of green for one approach till the next time the green starts.
  - It is denoted by C.

- **Interval**: Thus it indicates the change from one stage to another.
  - There are two types of intervals - change interval and clearance interval.
  - **Change interval** is also called the yellow time indicates the interval between the green and red signal indications for an approach.
  - **Clearance interval** is also called all red and is provided after each yellow interval indicating a period during which all signal faces show red and is used for clearing off the vehicles in the intersection.
- **Red interval**: It is the red indication for a particular movement or set of movements and is denoted by Ri.
- This is the **actual duration the red light of a traffic signal is turned on**.

- **Phase:** A phase is the green interval plus the change and clearance intervals that follow it.
  - Thus, during green interval, non conflicting movements are assigned into each phase.
  - It allows a set of movements to flow and safely halt the flow before the phase of another set of movements start.
- **Lost time:** It indicates the time during which the intersection is not effectively utilized for any movement.
  - For example, when the signal for an approach turns from red to green, the driver of the vehicle which is in the front of the queue, will take some time to perceive the signal (usually called as reaction time) and some time will be lost before vehicle actually moves and gains speed.

Phase design

The signal design procedure involves six major steps. They include the

1. phase design
2. determination of amber time and clearance time
3. determination of cycle length
4. apportioning of green time
5. pedestrian crossing requirements
6. the performance evaluation of the above design.

- The objective of phase design is **to separate the conflicting movements in an intersection into various phases, so that movements in a phase should have no conflicts.**
- If all the movements are to be separated with no conflicts, then a large number of phases are required.
- In such a situation, the objective is to design phases with minimum conflicts or with less severe conflicts.

- There is no precise methodology for the design of phases.
- This is often guided by the geometry of the intersection, flow pattern especially the turning movements, the relative magnitudes of flow.
- Therefore, a trial and error procedure is often adopted. However, phase design is very important because it affects the further design steps.
- Further, it is easier to change the cycle time and green time when ow pattern changes, where as a drastic change in the ow pattern may cause considerable confusion to the drivers.
- To illustrate various phase plan options, consider a four legged intersection with through traffic and right turns. **Left turn is ignored**. See figure 41:1.
- The **first issue is to decide how many phases are required**. It is possible to have two, three, four or even more number of phases

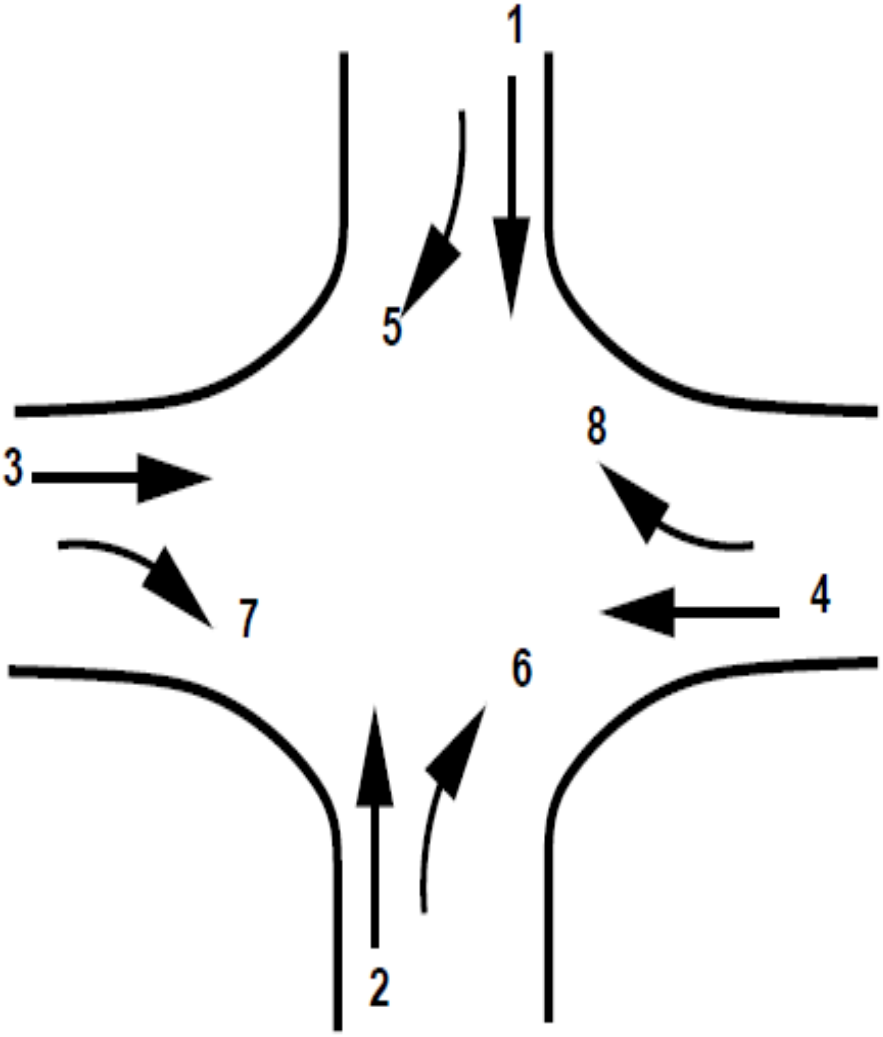


Figure 41:1: Four legged intersection

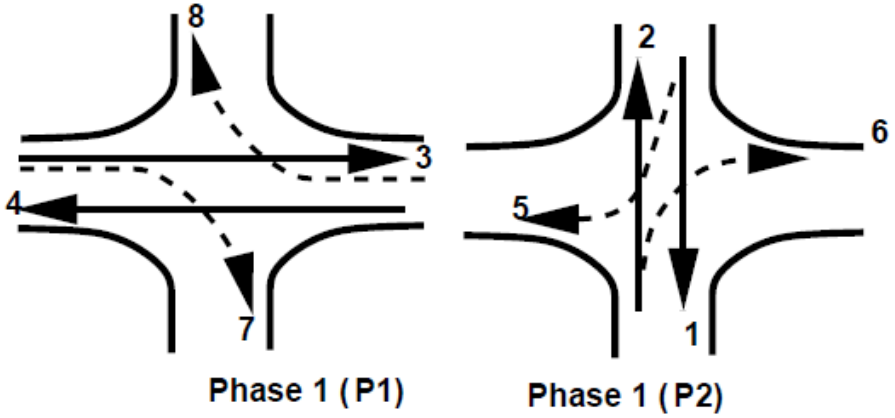


Figure 41:2: Two phase signal

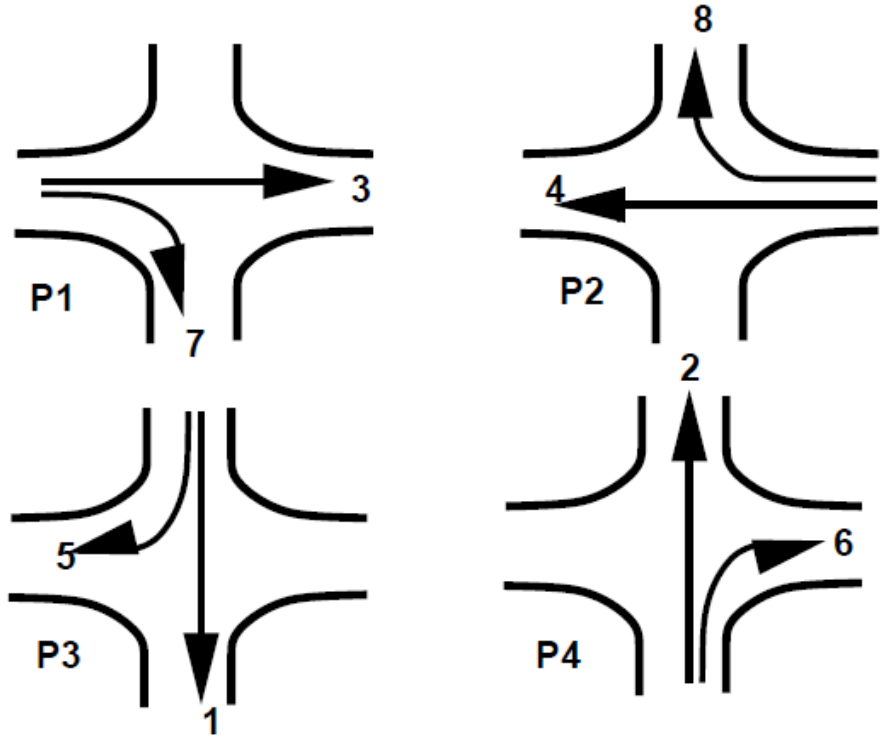


Figure 41:3: One way of providing four phase signals

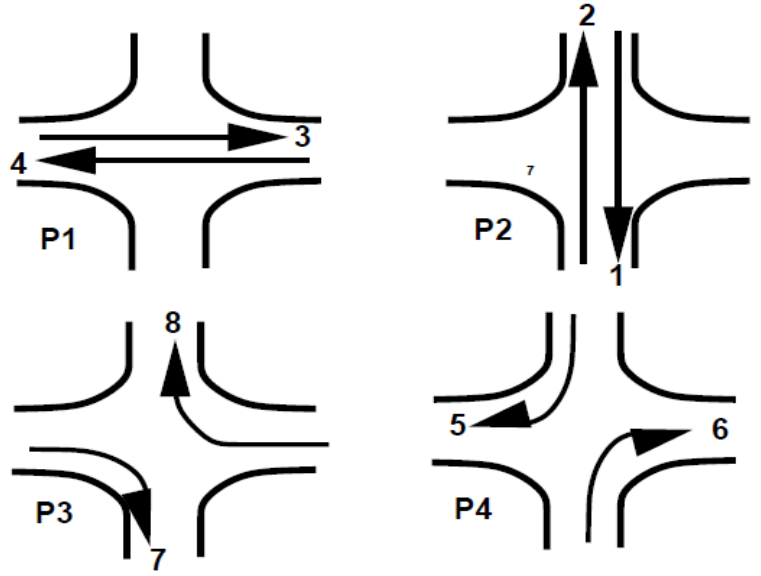


Figure 41:4: Second possible way of providing a four phase signal

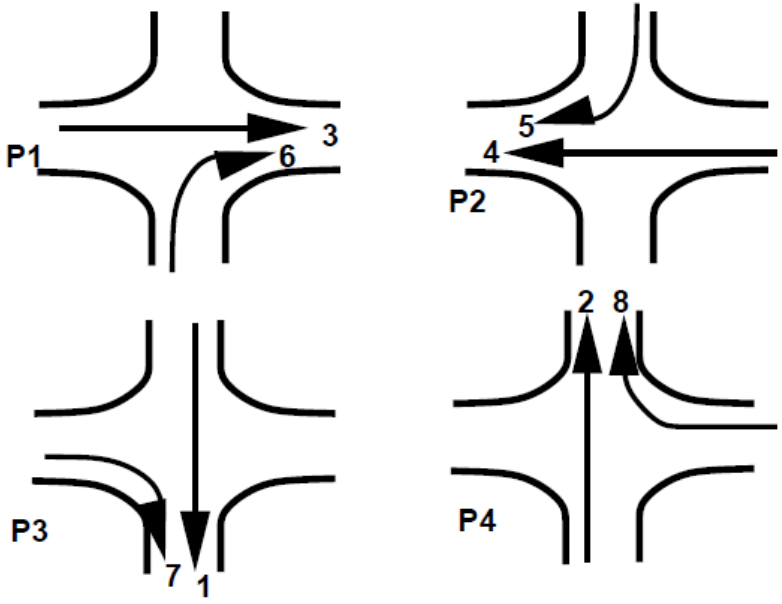


Figure 41:5: Third possible way of providing a four-phase signal

Signal Design Methods

Method Name	Key Features	Input Parameters	Advantages	Limitations
Webster's Method	**Most widely used method for fixed-time signals**	**- Traffic volume (v), saturation flow (s) - Lost time (L), number of phases (n)**	**- Simple - Provides optimal cycle time for minimal delay**	**- Assumes uniform arrivals - Not adaptive to fluctuating traffic**
IRC Method (Based on Webster)	**Adaptation of Webster's method with Indian conditions (IRC: SP: 41-1994)**	**- Similar to Webster - Adjusted parameters for Indian traffic**	**- Suitable for Indian traffic - Recommended by IRC**	**- Still assumes fixed-time signals**
Trial Cycle Method	Based on trial-and-error to optimize signal timings	- Traffic volumes - Trial cycle times - Green splits	- Practical for small intersections	- Time-consuming - Not precise for complex intersections
Equal Flow Ratio Method	Allocates green time proportional to critical flow ratios	- Critical flow ratios - Total cycle time	- Simple to implement	- Doesn't optimize delays - Less efficient for unbalanced flows
Shortest Cycle Method	Aims to minimize cycle time based on critical flows and saturation	- Critical flow ratios - Saturation flow - Lost time	- Reduces total delay	- May not consider pedestrian or minor phase needs
Maximum Efficiency Method	Maximizes the efficiency of the intersection (ratio of flow to capacity)	- Flow rates - Saturation flow - Cycle time	- Efficient use of capacity	- Requires accurate saturation flow estimation
Computer-Based Methods	Use of software/tools like TRANSYT, SYNCHRO for complex signal optimization	- Detailed traffic data - Signal phasing - Intersection geometry	- Accurate, handles complex networks	- Requires software and technical expertise

Webster's Method of Signal Design

Objective	To determine optimum cycle time for **fixed-time traffic signals** that minimizes delay at intersections.
Basic Assumptions	- Uniform traffic arrivals - Consistent flow during analysis period - Fixed-time signals - Two or more phases
IRC Adjustments	- Use actual traffic data from Indian roads - May assume standard lost time = 2 sec/phase - Saturation flow: 525 to 600 veh/hr/lane (IRC typical)

Key Formulae:

1. Optimum Cycle Time (C_0):

$$C_0 = (1.5L + 5) / (1 - Y)$$

Where:

C_0 = Optimum cycle time (in seconds)
L = Total lost time per cycle (in seconds)
Y = Sum of critical flow ratios (v/s) for all phases

2. Total Lost Time per Cycle (L): IN SEC

$$L = 2n + R$$

Where:
n = Number of phases
R = All red time or red-amber time

3. Critical Flow Ratio (Y):

$$Y = \Sigma (v_i / s_i)$$

$$Y = y1 + y2 + \ldots\ldots$$

$$y1 = v1 / s1$$

Where:
v_i = Actual traffic volume (veh/hr) of critical movement in phase i
s_i = Saturation flow (veh/hr) for that movement

4. Green Time Allocation (g_i):

$$g_i = (y_i / Y) \times (C_0 - L)$$

Where:

g_i = Effective green time for phase i

y_i = Flow ratio for phase i

5. Total Cycle Time (C):

$$C = C_0 = g_1 + g_2 + \ldots + g_n + L$$

6. Saturation Flow (s):

Typical value for Indian conditions: 525 to 600 vehicles/hour/lane

(as per IRC guidelines)

What is Rotary Intersection of Roads?

- **The rotary intersection of roads is also called as traffic rotary** which is nothing but enlarged intersection of roads where vehicles cross roads or change their direction without stopping.

- **All vehicles coming from different roads move in single direction around the central island and diverges into required exit.**

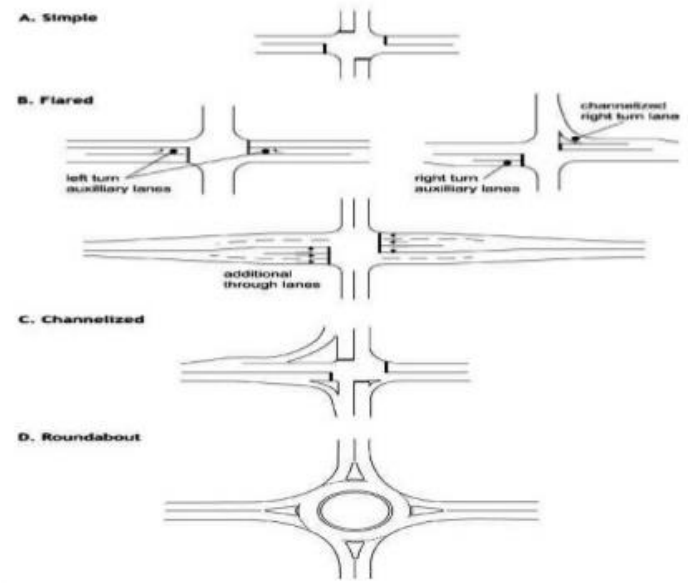




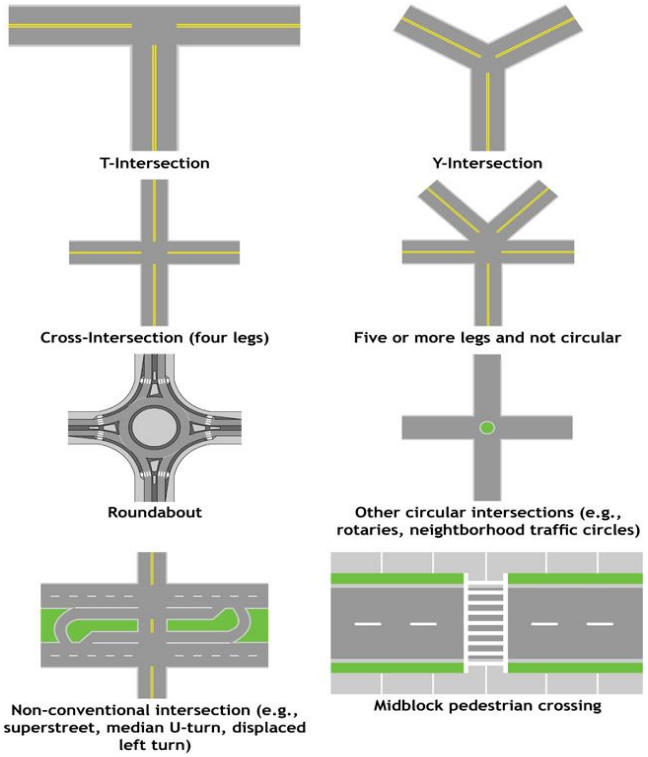
Road Intersection:

The places where two or more roads are arranged **to join or cross at same or different levels** are known as road intersection or road junctions.

1) **Intersection at Grade**: All the roadways join or cross at the same level. Un-channelized & channelized or rotary
2) **Grade Separated Intersection:** Separation in grade of the intersecting roads is achieved by providing a bridge. Over bridge or under pass.



Different types of at grade intersections

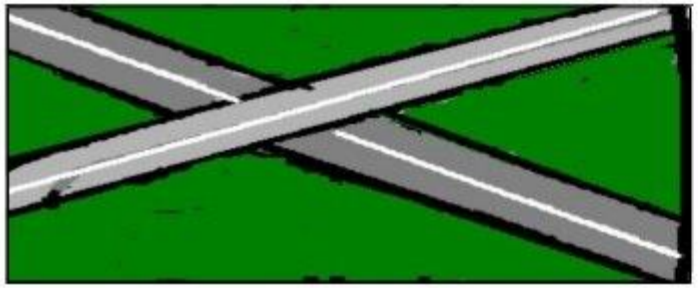


Types of Intersections



At Grade Intersection

(at same level)



Grade Separation

(Highway levels are not connected)



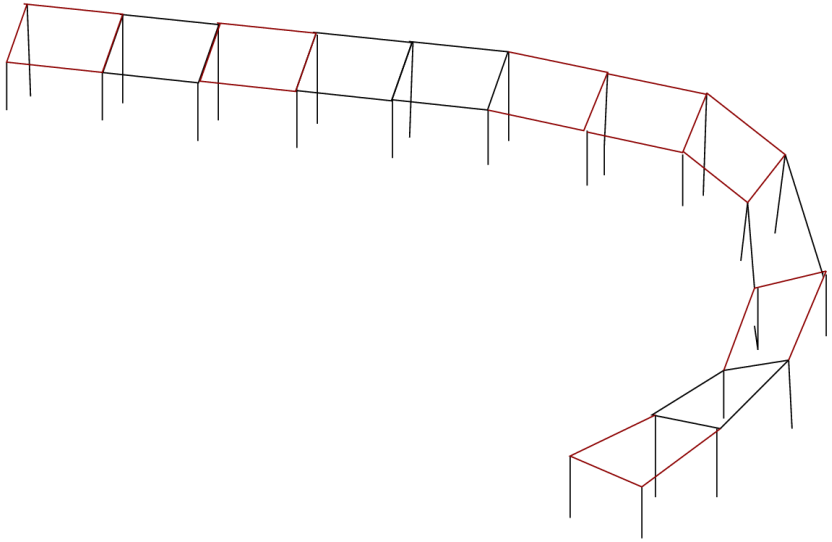
Interchange

(Highway levels are connected)

Classification of Grade Separated Intersection

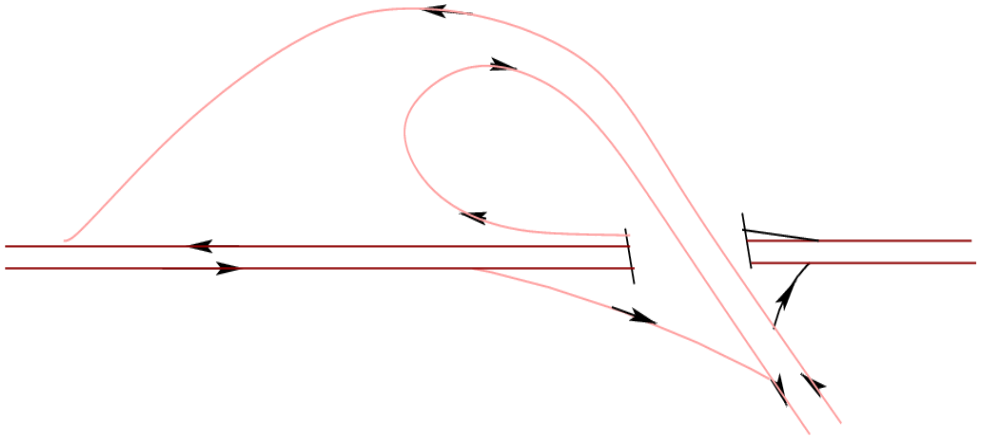
Overpass

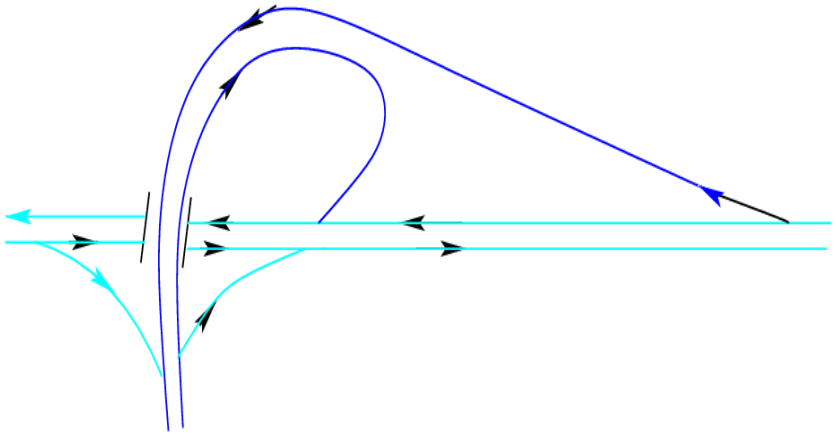
An overpass also known as a flyover, is a bridge, road, railway or similar structure that crosses over another road or railway



Trumpet Interchange

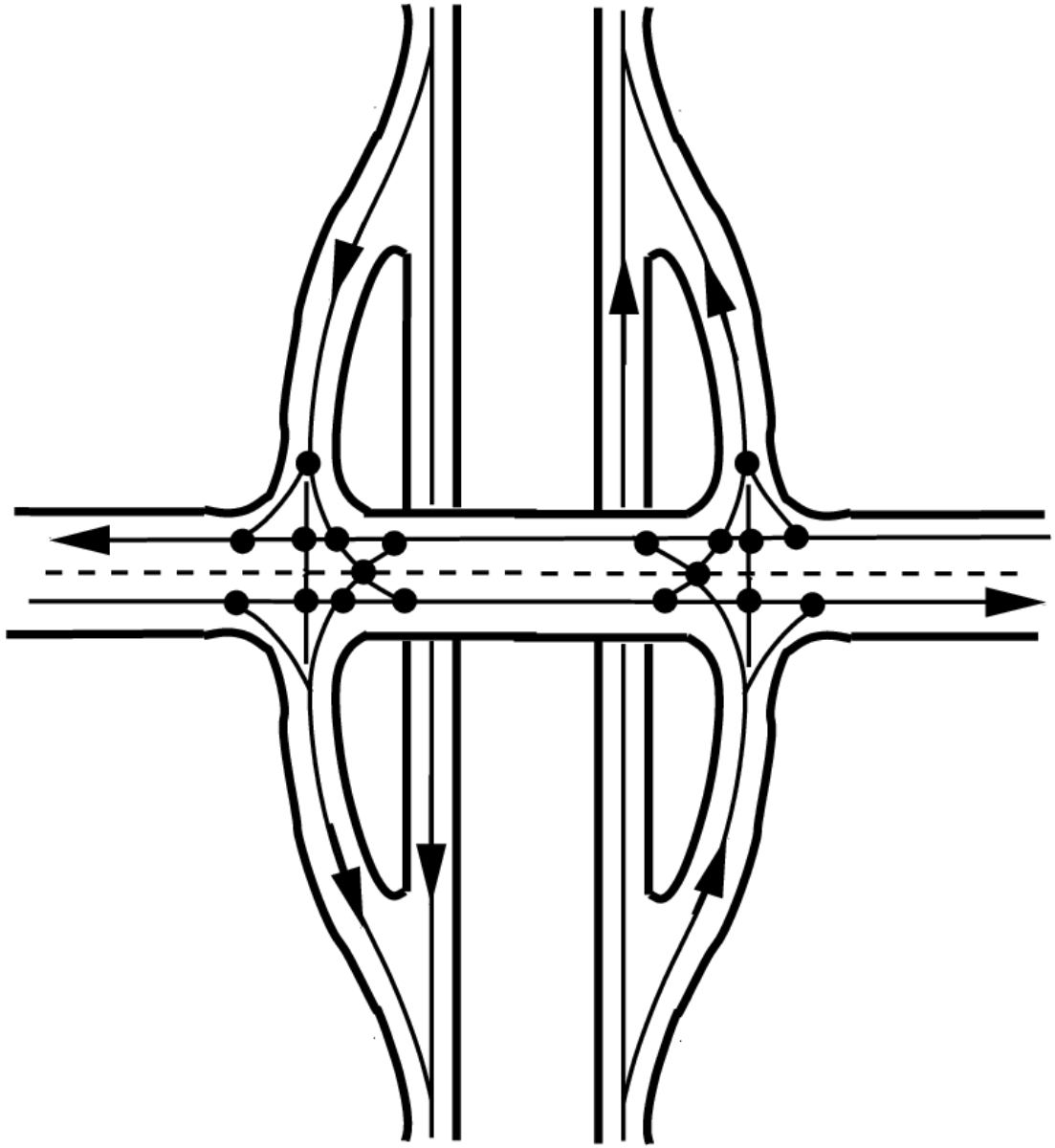
Trumpet interchanges have been used where one highway terminates at another highway





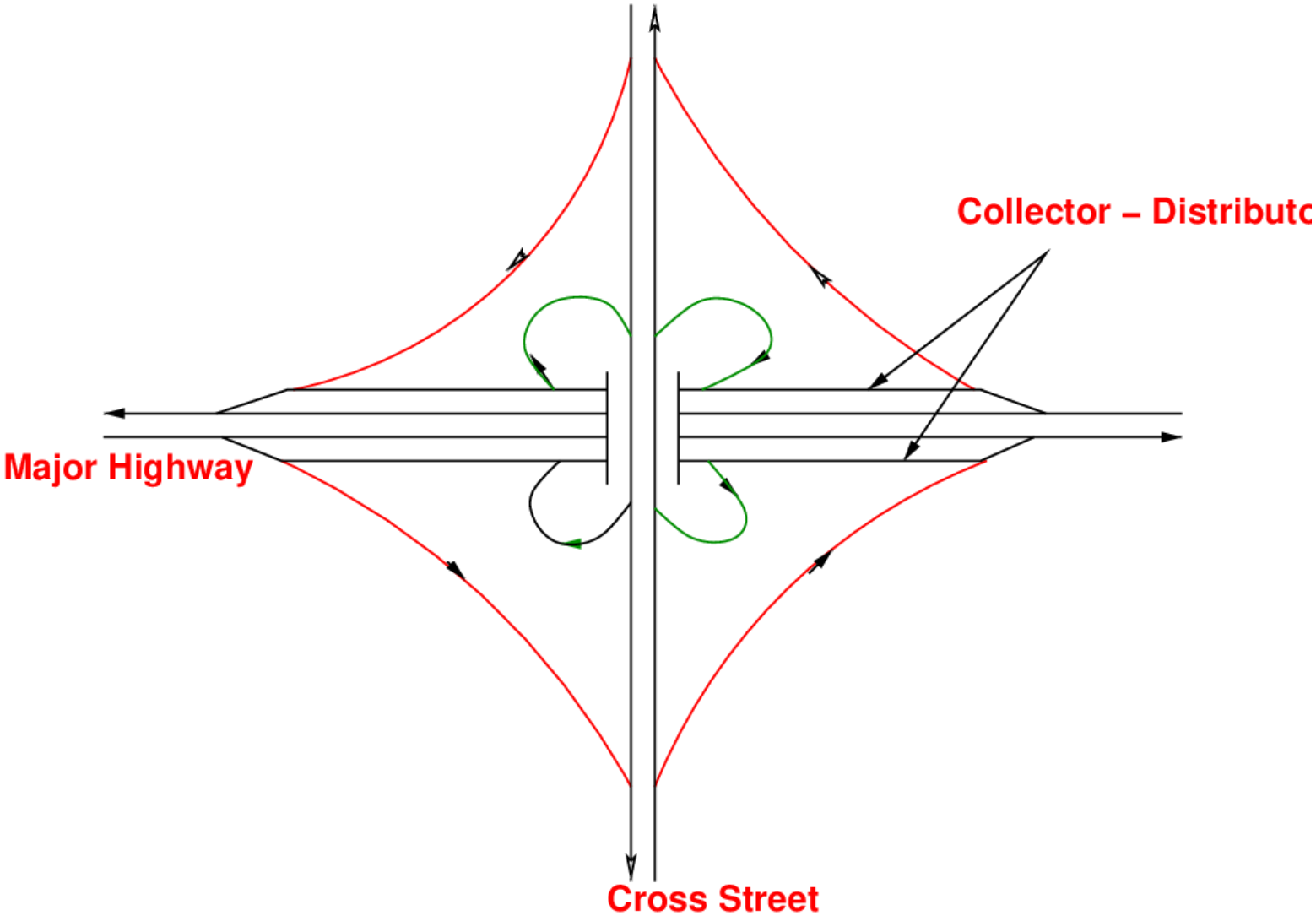
Diamond Interchange

The diamond Interchange is the simplest form of grade separated intersection between two roadways.



Cloverleaf Interchange

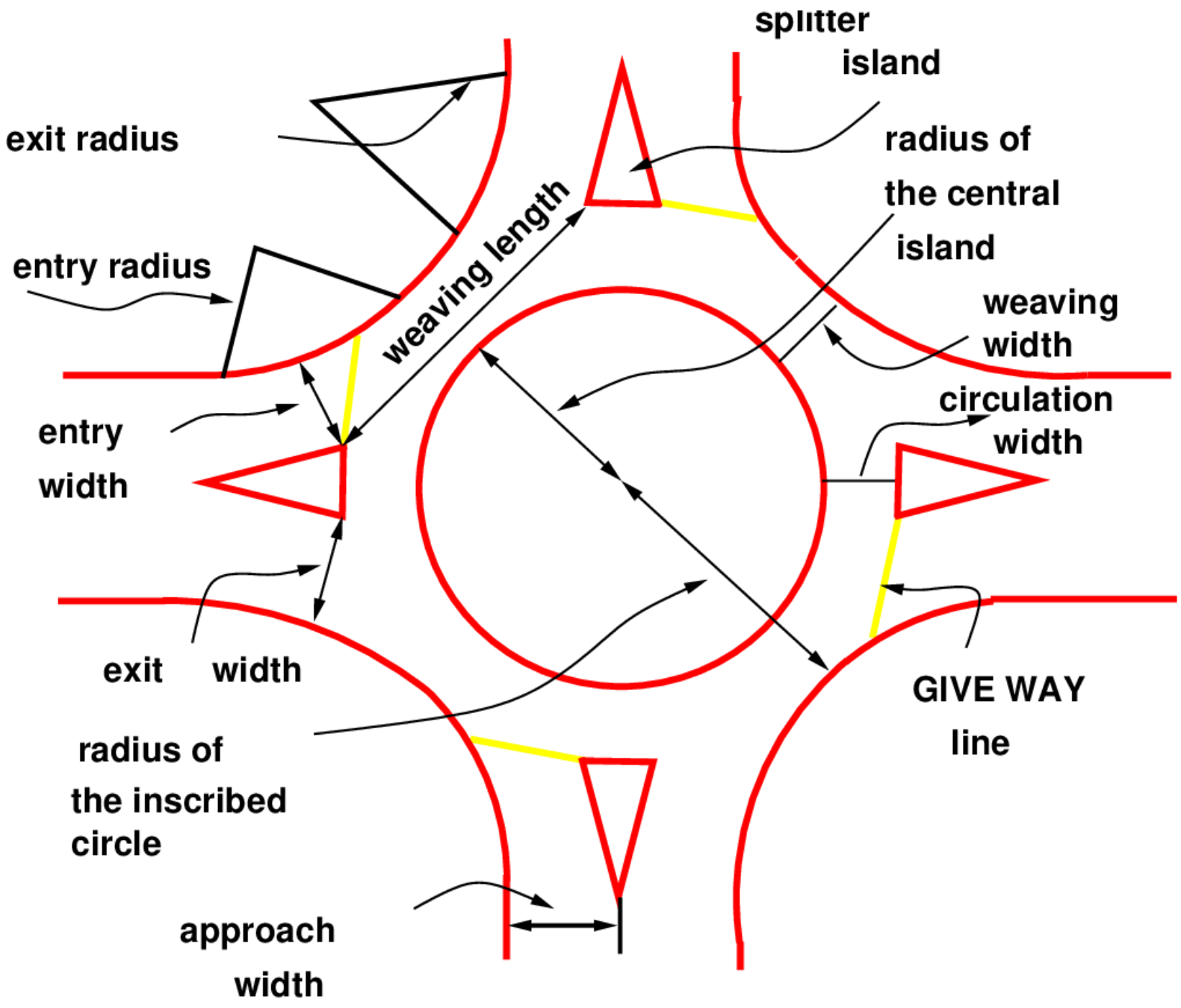
The full clover interchange eliminates all crossing movement conflicts by the use of weaving sections. This weaving section is a critical element of cloverleaf design. It replaces a crossing conflict with a merging, followed some distance farther by a diverging conflict.



Design of Rotary intersections

There are many factors to be considered while designing traffic rotary as follows:

- Design speed
- Shape of central island
- Radius of rotary roadway
- Weaving angle and weaving distance
- Width of carriageway at entry and exit
- Width of rotary roadway
- Curves at entrance and exit
- Capacity of rotary
- Channelizing islands
- Camber and super elevation
- Sight distance
- Lighting
- Traffic signs
- Pedestrian ways


splitter
island
exit radius
radius of
the central
island
entry radius
weaving length
weaving
width
circulation
width
entry
width
exit width
GIVE WAY
line
radius of
the inscribed
circle
approach
width

Design of Rotary intersections

There are many factors to be considered while designing traffic rotary as follows:

Design speed-

- At rotary intersections, the vehicle can be moved without any breaking.
- So, there is no chance to stop the vehicle while other vehicles can change their direction or cross each other without stopping.
- But, the speed of vehicular movement should be reduced.
- Standard highway speed is not applicable at rotary intersection.

Shape of central island-

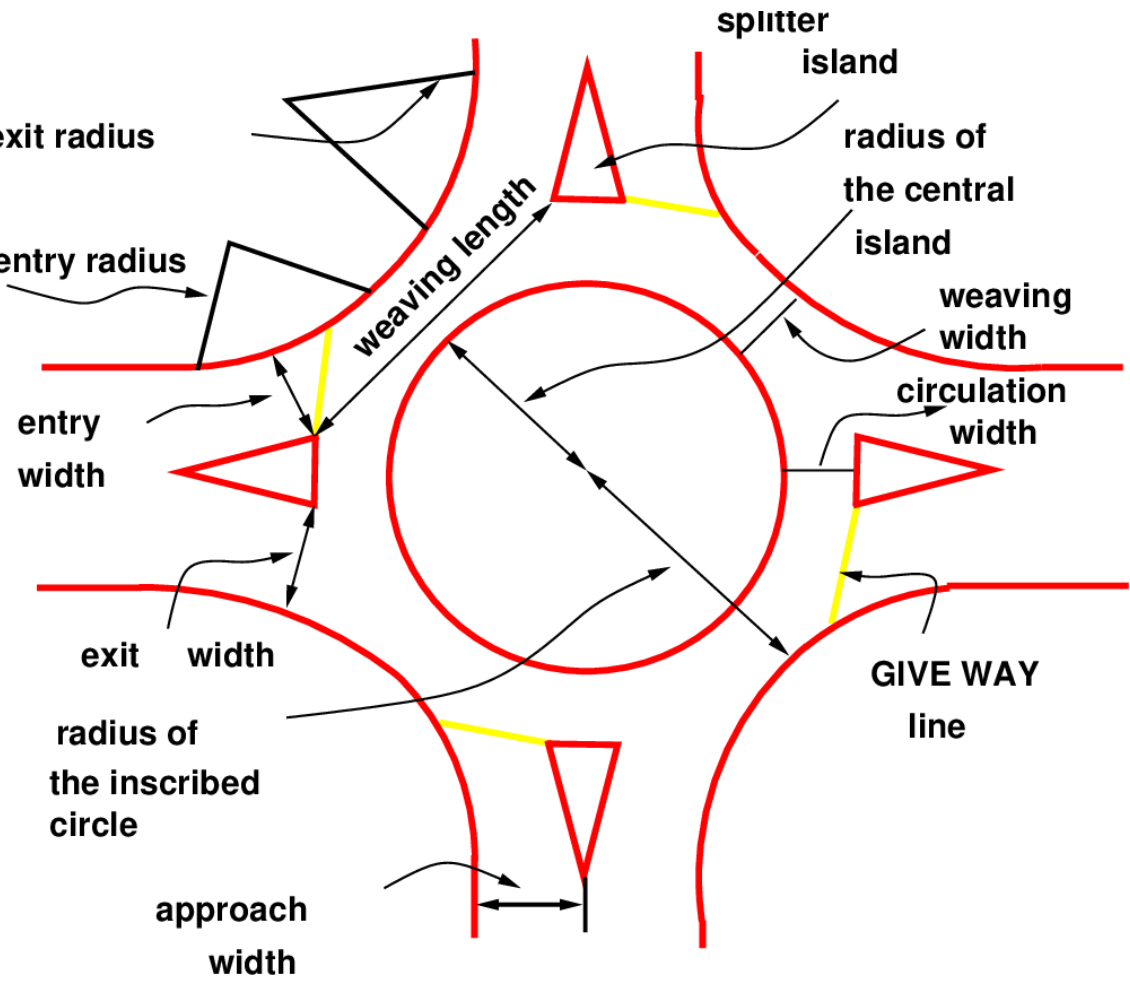
- The shape of Central Island provided for rotary intersection should not contain any corners.
- It should be formed by curves to allow the comfortable rotations around it.
- The shape is particularly dependent upon number of roads meeting at that particular junction.
- The shapes generally provided are circular, elliptical, turbine and tangential.

Radius of rotary roadway-

- The radius of roadway or pavement around the central island is dependent of shape of Central Island.
- If it is circular shape, radii are similar at all points and if it is elliptical or tangent radii is different at different points.
- The radius of rotary roadway should be designed by just considering the friction force and super elevation should be neglected.
- Normal radius of roadway in curves

$$R = \frac{V^2}{127(e+f)}$$

- But super elevation (e) is neglected i.e., e = 0 Hence, radius of rotary
- Where f = coefficient of friction = 0.43 to 0.47



Weaving angle and weaving distance-

- Weaving angle is the angle formed by paths of vehicle entering the rotary and other vehicle leaving the rotary at adjacent road.
- The exits of two vehicles may be different but they travel in same way for some short distance in the rotary section which is nothing but merging of vehicles and when the required exit is come two of the vehicles diverged into different directions.
- The length of which the two vehicles travel in same way is called as weave length.
- The weaving angle should be small but minimum of 15o is maintained.
- The weave length should be **at least four times the width of weaving section.**

Width of carriageway at entry and exit-

- The width of carriage way at entrance rand exits is dependent of volume of traffic in that particular region or area.
- But, the minimum width of 5.0 meters should be maintained for rotary intersections.

Width of rotary roadway

- Vehicles coming from all directions may meet at a time at least for shorter distance at rotary intersections.

$$W = \frac{(e_1 + e_2)}{2} + 3.5\,m$$

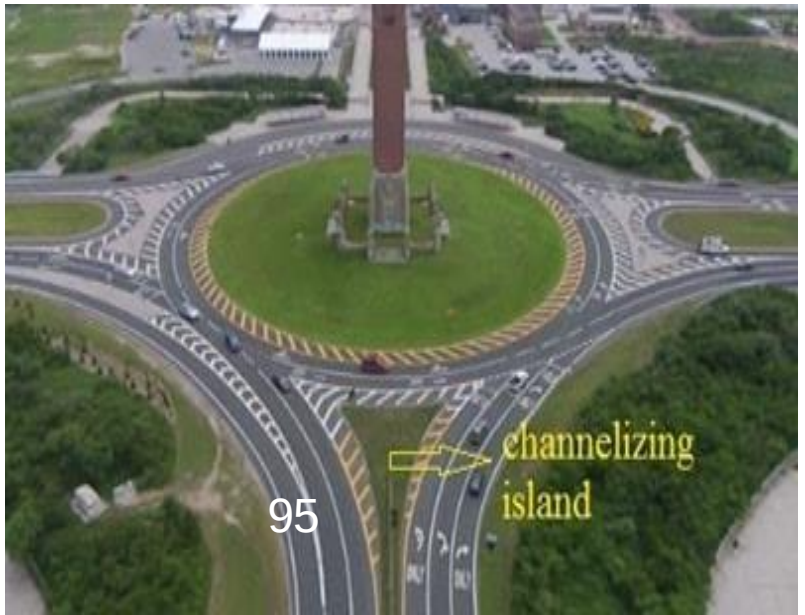
- The width should be equal to the effective width of weaving section.
- So, the width provided for rotary roadway should be as follows
- Where e1 = width at entrance e2 = width of non-weaving section

Curves at entrance and exit-

- Entrance and exit curve is nothing but a curve traced by the rear inner wheel of vehicle.
- Generally, at entrance the vehicle will slow down to design speed of rotary intersection so, at the entrance curve radius can be provided as same as radius of central island.
- Coming to exit curves, the vehicle accelerates at exits hence the radius of curve at exit should be greater than the radius of curve at entrance.

Channelizing islands

- Channelizing islands are provided at entrance or exit of road way to prevent the vehicle from undesirable weaving.



Capacity of the weaving section Cw (in PCU/hr)

- The capacity of rotary is derived from the below formula and it is mainly dependent upon capacity of individual weaving section.

$$C_w = \frac{280 \cdot w \cdot \left(1 + \frac{e}{w}\right) \cdot \left(1 - \frac{p}{3}\right)}{1 + \frac{w}{l}}$$

- Where

w = Width of the weaving section (in meters)

e = Average entry width (in meters)

L =Length of the weaving section (in meters)

p = Proportion of weaving traffic to total traffic =(b+c)/(a+b+c+d) = (0.4 < p < 1.0)

=(weaving traffic/total Traffic)

a = left turning traffic moving along left extreme lane

b = weaving traffic turning toward right while entering the rotary

c = weaving traffic turning toward left while leaving the rotary

d = right turning traffic moving along right extreme lane

Camber and super elevation-

- The super elevation for rotary roadways is neglected.
- But, here if the vehicle is changing its direction to its opposite side it will travel around the central island and changes the direction.
- While changing, the vehicle may over turn or slip, to overcome this, minimum cross slope is provided which is nothing but camber.
- This camber acts as super elevation in case of rotary roadways.

Sight distance

- The sight distance provided at rotary intersections should be as higher as possible and in no case the value must be less than the stopping sight distance.

Lighting-

- The edge of Central Island should be installed with lights which is mandatory.
- Additional lights may also be provided at the kerbs if the diameter of Central Island is more than 60 m.
- Sometimes, entrance and exit curves can also be provided with lights.

Traffic signs-

- Traffic signs should be installed on approaching roads to indicate the presence of rotary intersection ahead to the roadway users.
- Kerbs at rotary intersections should be coated with black and white strips to improve visibility.
- Traffic signals should be placed 1 meter above road level to indicate the direction of exit.

Parking Facility:

Vehicles do not keep moving along the road during entire 24 hours of the day.

Problems of parking is becoming more & more acute day by day.

I. **On Street or Kerb Parking**: Vehicles are parked along the kerb. Parallel parking & Angle parking.

II. **Off Street Parking:** If perking demand is more, kerb parking can not be permitted in views of traffic congestion. **Parking facility at separate place away from the road is known as off street parking.** It should be provided at the nearest locations depending upon the availability of space. Surface Parking Lots & Multi Storeyed Parking Garages.

TYPES OF KERB PARKING



- **Parallel Parking**

  Vehicles are parked parallel to kerb.

  It is adopted when width of street is limited.



- **Angle Parking**

  It is more convenient for the motorists than the parallel parking

  45° angle parking is the best and most convenient.

  - 30° angle parking
  - 60° angle parking
  - 90° angle parking









Off Street Parkings are of two types:

1) Surface Parking Lots &
2) Multi Storeyed Parking Garages**.**





Parking Space

Identify The Purpose of Lot

- Parking lot size
- Pavement thickness
- Parking space angles

Sizing and Spacing

- Number of spaces
- Parking slot size
- Accessibility Requirements
- Uses for extra space

Parking Layout Design

- Space efficiency
- Traffic signage
- Painted asphalt markings
- Concrete parking barriers
- Drainage provisions
- Access ramps
- Lighting design

Other Design Elements

- Landscaping design
- Barrier gates
- Emergency call stations

ParkCAD™ is an advanced CAD software that gives users the right tools to design parking layouts quicker and smarter than traditional CAD methods.

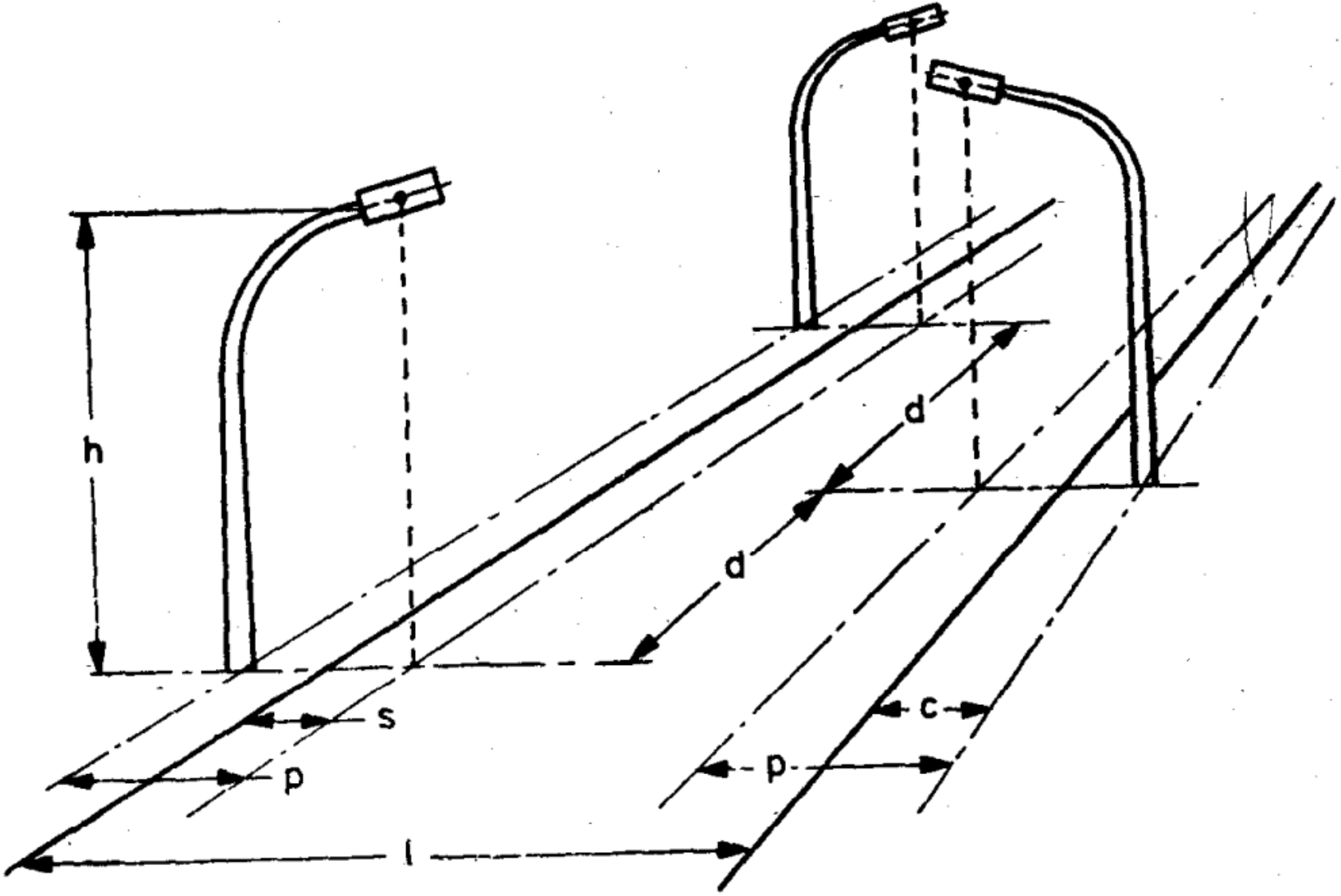
https://www.educative.io/courses/grokking-the-object-oriented-design-interview/gxM3gRxmr8Z

Parking requirements

- There are some minimum parking requirements for different types of building.
- For residential plot area less than 300 sq.m require only community parking space.
- For residential plot area from 500 to 1000 sq.m, minimum one-fourth of the open area should be reserved for parking.
- Offices may require at least one space for every 70 sq.m as parking area.
- One parking space is enough for 10 seats in a restaurant where as theatres and cinema halls need to keep only 1 parking space for 20 seats.
- Thus, the parking requirements are different for different land use zones.

Highway Lighting

- Highway lighting consists of **providing illumination through lights** in the vicinity of a highway/road/street.
- It plays a major part in ensuring of **highway safety during night driving** and includes the aspect of roadside amenities for the convenience of the road- users, and preserving the aesthetics of the highway and its vicinity during the nights.
- In addition, highway lighting is **important to provide security to the road-users** during nights, especially to pedestrians.
- Accidents are known to occur more during nights than during the day; this is attributed to poor visibility of potential hazards at night time and lack of proper lighting for the road or highway.



h = mounting height of luminaires
d = spacing
l = width of the carriageway
p = outreach
s = overhang
c = clearance

FIG. 1 SITING OF LUMINAIRES: CHARACTERISTIC DIMENSIONS

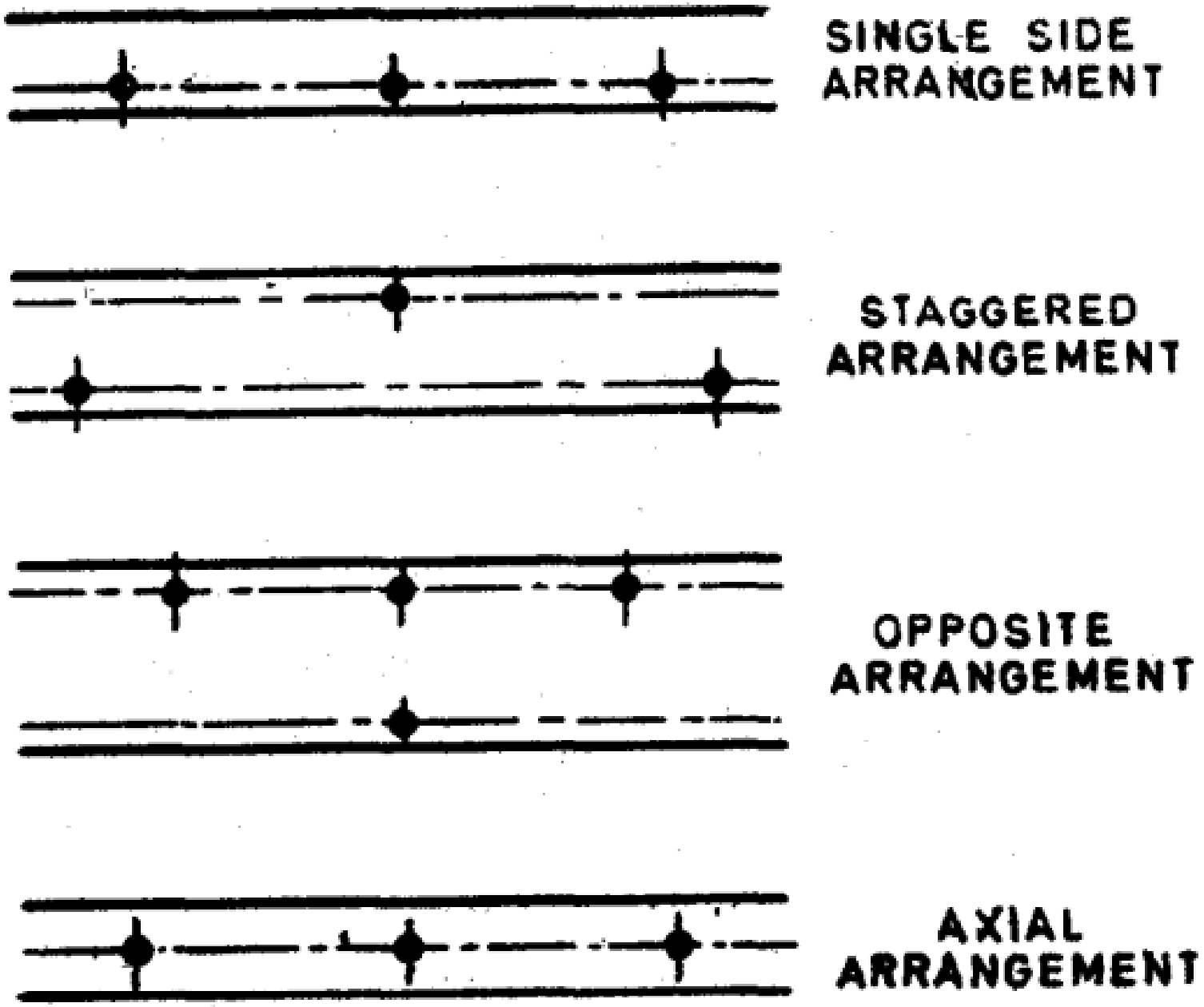


FIG. 2 BASIC ARRANGEMENTS IN PUBLIC LIGHTING

Illuminance (Luminance) Levels

As per IRC, minimum average illuminance for different road types is:

Road Type	Average Illuminance (Lux)
Expressways / Urban Freeways	30 - 50 lux
Major urban arterials	20 - 30 lux
Minor urban arterials	10 - 20 lux
Residential streets	4 - 10 lux
Intersections	1.5 × adjacent road level

1. Objectives of Highway Lighting	- Improve night visibility - Reduce accidents - Improve traffic efficiency - Assist in route guidance - Enhance safety for pedestrians
2. Types of Lighting Systems	Based on Mounting Height: • Low-mounted (up to 3 m) • Medium-mounted (3–9 m) • High-mounted (10–15 m) Based on Distribution: • Symmetrical • Asymmetrical • Cut-off
3. Types of Lamps	- High-pressure sodium (HPSV): Efficient, yellow light - Metal halide: White light, good color rendering - LED: Long life, energy efficient - Mercury vapor: Obsolete, low efficiency
4. Design Parameters	- Spacing of lamps - Mounting height - Overhang from edge - Wattage - Road width & traffic volume
6. Recommended Illuminance (E)	- Expressways / Major roads: 20–30 lux - Urban streets: 10–20 lux - Residential streets: 4–8 lux - Footpaths: 2–4 lux
7. Recommended Mounting Height	- Residential: 6–9 m - Arterial roads: 9–12 m - Expressways: 12–15 m
8. Glare Control	- Use cut-off fixtures - Maintain correct tilt angle - Use anti-glare shields
9. Pole Placement	- Located beyond the clear zone (≥1.5 m from edge of pavement) - Double-sided staggered or median mounting for wide roads

Planning and Administration

Traffic Planning:

- Traffic planning comprises **evaluation, designing, assessment and management** of transportation.
- It involves **recognition of problems**, making out **solutions** and planning.
- Multilane roads in a Highway need **systematic traffic operation** to avoid accidents and maintain road safety.
- Transportation planning **also involves study of transport demand, economic evaluation, analyzing of urban and interurban transport terminals** and consideration of issues related to environment.

1. LAND USE PLANNING AND ROAD TRAFFIC
2. TRAFFIC FORECAST
3. ACCESS CONTROL ON HIGHWAYS
4. ROAD SAFETY

1. LAND USE PLANNING AND ROAD TRAFFIC

Objective	- Integrate road network with land development - Promote balanced growth - Minimize travel demand
Importance	- Efficient zoning reduces traffic congestion - Promotes accessibility - Encourages public transport use
Stages / Approach	- Land Use Survey (residential, commercial, etc.) - Trip Generation Analysis - Zoning & Road Hierarchy Design (arterials, sub-arterials, collectors) - Integration with Master Plan (urban/local planning authorities) - Traffic Impact Assessment (TIA) for new developments
IRC Reference	IRC:SP:88-2010 – Guidelines on Road Traffic and Land Use Planning

2. TRAFFIC FORECASTING

Objective	- Estimate future traffic volume - Aid in long-term infrastructure design - Prevent over/under design
Importance	- Ensures optimal capacity design - Helps plan for future modes (BRT, metro) - Supports investment decisions
Stages / Methods	1. Data Collection (current volume, O-D surveys) 2. Trend Analysis (growth rate from past years) 3. Modeling Techniques: • Linear Growth • Travel Demand Models (4-stage model: trip generation → distribution → modal split → assignment) 4. Forecast Horizon: 10, 15, or 30 years 5. Validation and Sensitivity Checks
IRC Reference	IRC:108-2015 – Traffic Forecasting & Capacity Estimation

3. ACCESS CONTROL ON HIGHWAYS

Objective	- Improve safety and efficiency of highways - Eliminate conflict points - Maintain uninterrupted flow
Importance	- Reduces accidents - Enhances travel speed - Helps in expressway development - Reduces delays at intersections
Types of Access Control	- Full Access Control (expressways – no at-grade intersections) - Partial Access Control (NH with limited median openings, service roads) - No Access Control (rural roads, low volume roads)
Stages / Measures	- Use of grade separators (flyovers, underpasses) - Fencing and boundary treatments - Service roads for local access - Designated entry/exit ramps - Enforcement through signages & laws
IRC Reference	IRC:SP:99-2013 – Guidelines for Access Control on National Highways IRC:SP:87-2013 – Design of Access-Controlled Highways

4. ROAD SAFETY ISSUES

Objective	- Reduce road traffic crashes - Ensure safe mobility for all users - Identify and mitigate accident-prone areas
Importance	- Saves lives and property - Enhances public confidence in road use - Essential for sustainable transportation
Key Issues	- Over-speeding, signal violations - Poor road geometry - Lack of signage - Unsafe pedestrian movement - Mixed traffic conditions
Stages / Measures	1. Accident Data Collection & Analysis 2. Identification of Black Spots 3. Engineering Measures (speed calming, signage) 4. Education & Awareness (road safety campaigns) 5. Enforcement (traffic police, automated systems) 6. Emergency Response Planning
IRC Reference	IRC:SP:88-2010 – Road Safety Audit Manual IRC:SP:55-2014 – Guidelines on Safety of Vulnerable Road Users

Highway Administration:

Purpose

- To plan, construct, operate, and maintain road infrastructure efficiently.
- Ensure safety, legal compliance, and effective policy implementation.
- Coordinate between various governmental and non-governmental agencies.

Level	Agencies / Authorities
Central Government	- Ministry of Road Transport & Highways (MoRTH) - National Highways Authority of India (NHAI) - Indian Roads Congress (IRC) - Central Road Research Institute (CRRI)
State Government	- Public Works Department (PWD) - State Road Development Corporations (e.g., MSRDC) - State Transport Departments
Local Bodies	- Municipal Corporations - Panchayats - Urban Local Bodies (ULBs) - Development Authorities

Road Accidents and Safety:

Classification, Causes, Mitigation and Control Measures, Aspects of Safety in Usage of Roads, Type and Design of anti-crash barriers

- Introduction to Intelligent Transport Systems (ITS).

Road Accidents and Safety:

90% OF ALL ROAD ACCIDENTS ARE LINKED TO HUMAN ERROR

THE BEHAVIOUR OF ROAD USERS IS THE AREA WITH BY FAR THE BIGGEST POTENTIAL FOR IMPROVING ROAD SAFETY



IN 30% OF FATAL ACCIDENTS SPEEDING IS THE MAIN FACTOR



DISTRACTION CAUSES 10-30% OF ROAD DEATHS



25% OF ALL ROAD FATALITIES IN EUROPE ARE ALCOHOL-RELATED

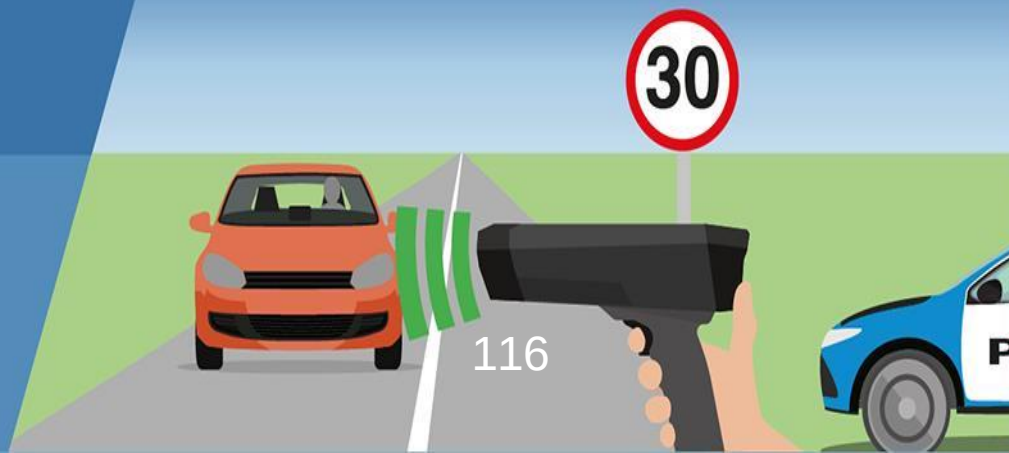


ABOUT 65% OF FATAL ACCIDENTS ARE CAUSED BY VIOLATIONS OF TRAFFIC RULES

EDUCATION AND TRAINING ARE CRUCIAL IN INSTILLING APPROPRIATE BEHAVIOUR AND ATTITUDES IN ROAD USERS



CRACKING DOWN ON TRAFFIC OFFENCES WILL MAKE A DIFFERENCE



Classification of Road Accidents (As per IRC)

Category	Sub-Classification
A. Based on Severity	- Fatal accident: At least one death - Major injury: Serious injuries (hospitalized >24 hrs) - Minor injury: Injuries but not serious - Non-injury (Property Damage Only)
B. Based on Type of Collision	- Rear-end collision - Head-on collision - Side swipe - Right-angle (T-bone) - Run-off-road - Hit pedestrian/cyclist
C. Based on Location	- Intersection - Mid-block - Curve - Bridge approaches - Access/exit ramps
D. Based on Time of Occurrence	- Day / Night - Peak hours / Off-peak
E. Based on Road User Involved	- Pedestrian - Cyclist - Motorcyclist - Car - Truck / Bus

Causes of Road Accidents (As per IRC:SP:88-2010 and SP:55-2014)

Category	Cause
A. Human Factors	Over-speeding
	Drunken driving
	Fatigue and drowsiness
	Distraction
	Violations of traffic rules
	Lack of skill / training
	Aggressive behavior
B. Vehicle Factors	Mechanical failure
	Improper maintenance
	Overloading
C. Road Environment Factors	Poor geometric design
	Lack of signage / markings
	Slippery / uneven surfaces
	Inadequate lighting
	Improper drainage
	Construction zones
D. Environmental Factors	Fog / low visibility
	Rain / snow
	Bright sun glare

Hierarchy of Controls to Mitigate Traffic Risks

Use the hierarchy of controls below to manage interaction between road users and the work zone. Most work zones will require a combination of these controls.

Elimination/Substitution

Includes the use of detours, alternate routes or barriers.

Engineering Controls

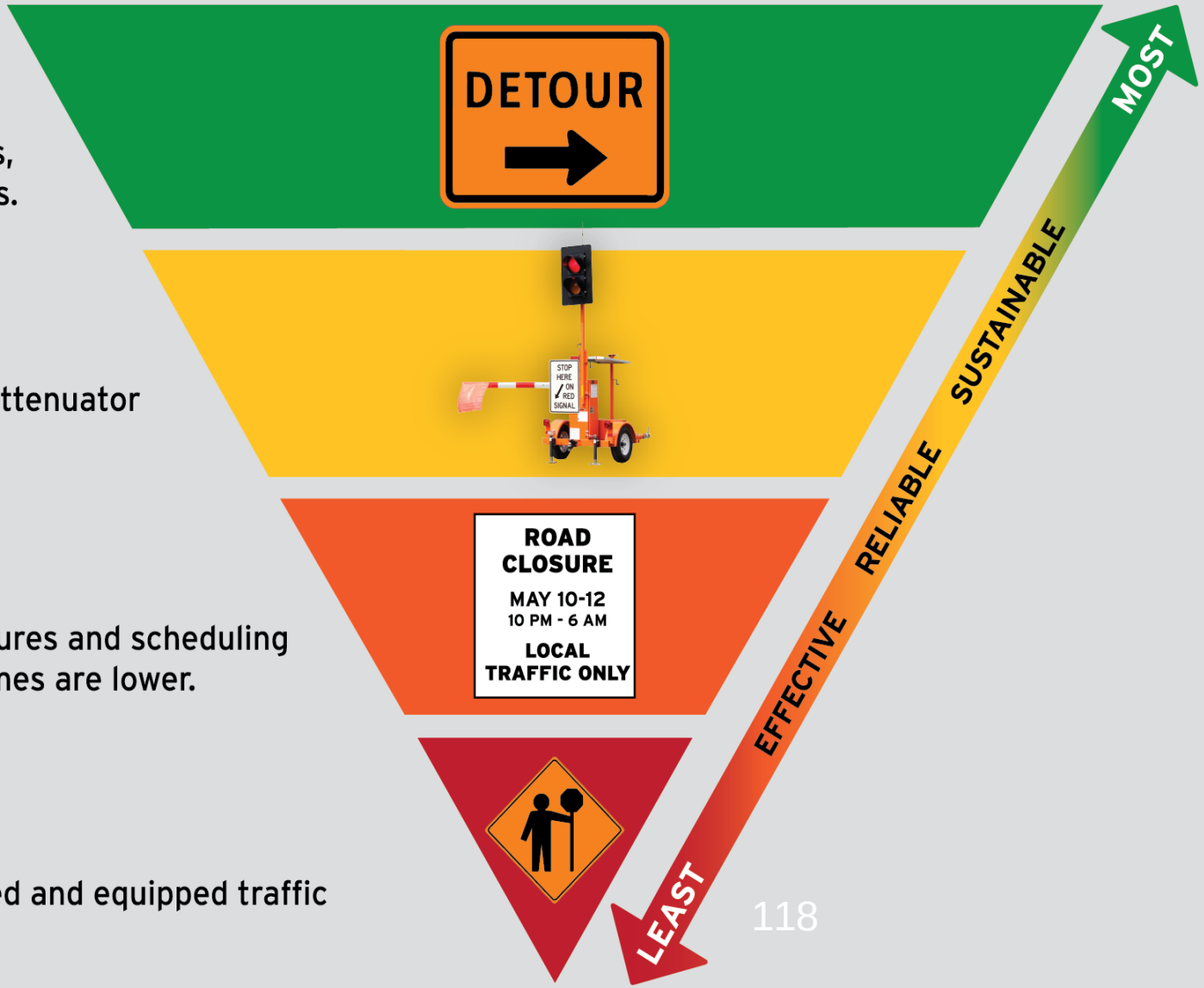
Includes the use of crash attenuator or traffic control devices.

Administrative Controls

Includes the use of procedures and scheduling work for when traffic volumes are lower.

Traffic Control Persons

Use of appropriately trained and equipped traffic control personnel.



Category	Measure	Description / Example
1. Engineering Measures	Road Design Improvement	Proper alignment, sight distance, lane width, curve design
	Speed Control Devices	Speed bumps, rumble strips, chicanes, posted speed limits
	Signage and Markings	Clear road signs, lane markings, pedestrian crossings
	Lighting and Visibility	Streetlights at intersections, reflectors, retro-reflective signs
	Pedestrian Facilities	Footpaths, zebra crossings, footbridges, pedestrian refuge islands
	Crash Barriers	Guardrails, median barriers, crash cushions
	Safer Intersections & Roundabouts	Grade separation, signalization, rotary design
2. Enforcement Measures	Traffic Law Enforcement	Speed guns, red light cameras, traffic patrols
	Penalties & Fines	Challans for over-speeding, drunk driving, and mobile phone usage
	Public Awareness Campaigns	Safety messages, community events, road safety week
	Overloading Control	Axle load checks, vehicle weighing stations
	Driver Testing & Licensing	Strict driving tests, periodic license renewal for commercial drivers
3. Educational Measures	Public Education Programs	School campaigns, media awareness, posters, seminars
	Driver Training	Pre-license training, refresher courses for commercial drivers
	Awareness for Vulnerable Users	Campaigns for pedestrians, cyclists, helmet and seatbelt awareness
4. Technological Measures	Intelligent Traffic Systems (ITS)	Adaptive signals, CCTV surveillance, traffic flow monitoring
	Driver Assistance Systems	ABS, ESC, AEB, lane assist, adaptive cruise control
	Collision Avoidance Tech	Proximity sensors, automatic braking systems
5. Road Safety Audits	Safety Audits (RSA)	Conducted at planning, design, construction, and operation stages
	Black Spot Identification	Mapping high accident areas and improving them
	Post-Accident Investigations	Data collection, cause analysis, corrective action planning
6. Emergency Measures	Emergency Response Systems	Ambulance availability, emergency hotlines, quick response teams
	First-Aid Training	First aid kits in vehicles, driver and responder training
	Trauma Care Facilities	Establishing trauma care centers near accident-prone corridors

Aspects of Safety in Usage of Roads



Aspect	Description	Key IRC Guidelines
1. Road User Behavior	Promoting safe behavior among all users including drivers, pedestrians, and cyclists.	IRC:SP:55-2014
2. Speed Management	Ensuring appropriate speed limits based on road classification, geometry, and surroundings.	IRC:SP:88-2010, IRC:99
3. Road Design & Geometry	Safe alignment, gradient, visibility, and curvature to reduce crashes and enhance maneuverability.	IRC:73 (Geometric Design), IRC:86, IRC:35
4. Signage and Marking	Clear, visible and well-maintained traffic signs and pavement markings for guidance and warnings.	IRC:67 (Signage), IRC:35
5. Roadside Environment	Managing trees, poles, slopes, and clear zones to reduce collision severity in case of runoff.	IRC:SP:21, IRC:SP:87
6. Pedestrian Safety	Safe crossing, sidewalks, refuge islands, and proper lighting for non-motorized users.	IRC:103, IRC:SP:88
7. Access Control	Restricting random access to highways to maintain flow and minimize conflict points.	IRC:SP:73, IRC:SP:84
8. Vehicle Safety Features	Encouraging use of safety-enhanced vehicles with features like ABS, seatbelts, and airbags.	MoRTH standards, indirectly covered by IRC
9. Visibility and Lighting	Adequate night-time lighting especially at junctions, crossings, and accident-prone zones.	IRC:SP:73, IRC:SP:88
10. Enforcement Infrastructure	Use of surveillance systems, cameras, and proper signboards for speed and rule enforcement.	IRC:SP:55, MoRTH Safety Guidelines
11. Road Safety Audit (RSA)	Safety evaluation at design, construction, and operational stages to identify and reduce risks.	IRC:SP:88-2010 (Full Manual on RSA)
13. Safety during Construction	Work zone traffic management, barriers, signs, and speed controls to protect workers and users.	IRC:SP:55-2014, MoRTH Safety Code
14. Emergency Facilities	Provision for lay-byes, emergency call boxes, ambulance access, and trauma care centers.	IRC:SP:55, IRC:SP:84
15. Public Awareness	Education campaigns for safe road usage by all categories of users.	MoRTH / IRC outreach guidelines

Type and Design of anti-crash barriers

Type of Barrier	Description	Application / Use	Design Criteria	
1. W-Beam Crash Barrier	Steel rail barrier with W-shaped corrugated section mounted on steel posts.	Roadside or median barriers on highways and expressways	- Post spacing: 2 m - Rail height: 700 mm from ground - Embedded depth: 1.2 m - End anchorage and crashworthiness tested to TL-3 standard	
2. Thrie-Beam Barrier	Triple-wave beam for greater impact resistance than W-beam.	Critical zones: bridges, steep slopes, high-speed roads	Similar design as W-beam but with increased depth and strength - Better redirection and energy absorption	
3. Concrete Barrier (New Jersey Type / F-Shape)	Precast or cast-in-situ rigid barrier with a sloped face.	Median or roadside on high-speed highways, bridges	- Height: 810 mm (standard) - No deflection (rigid) - Reduces vehicle cross-over - Suitable for narrow medians	

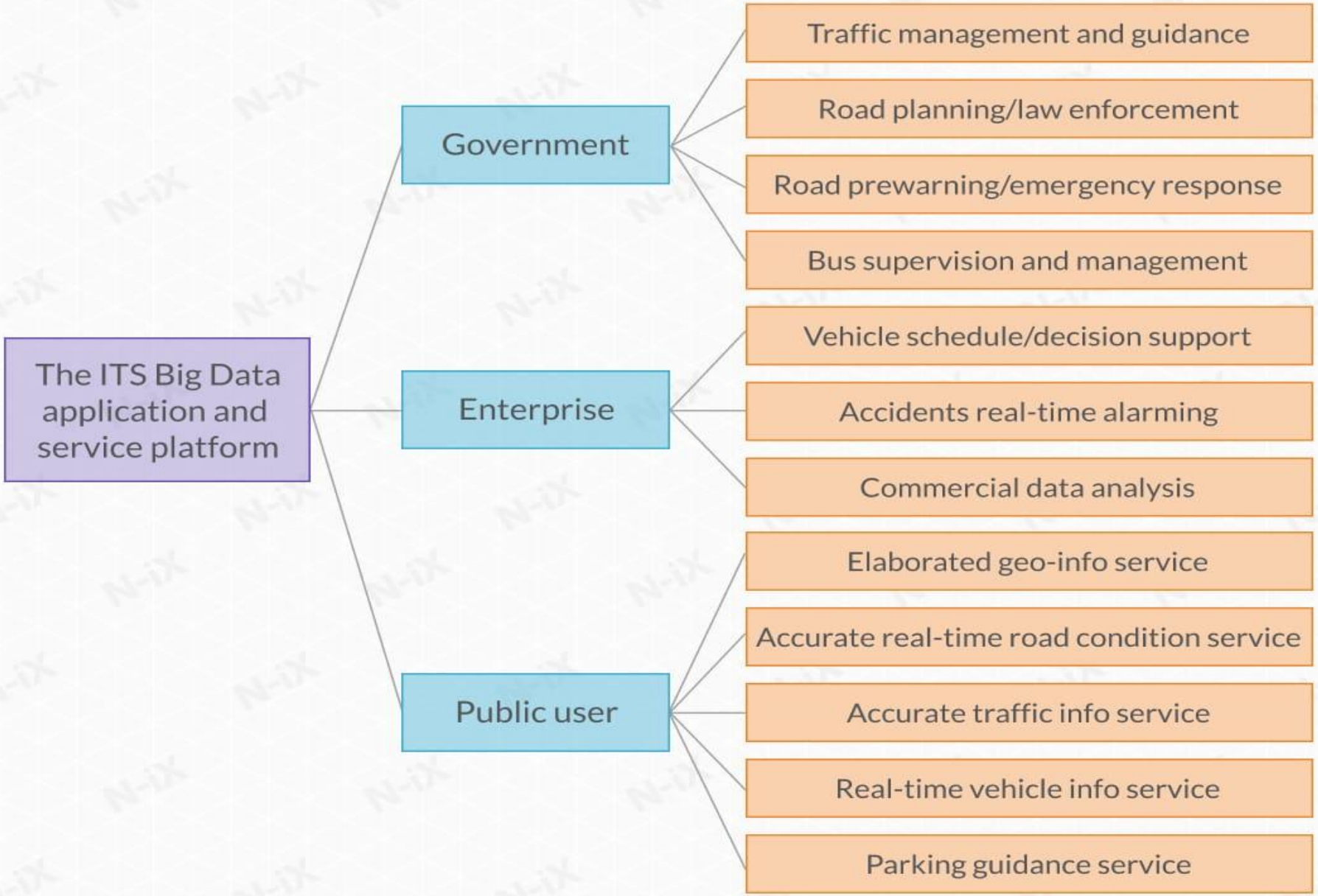
Type of Barrier	Description	Application / Use	Design Criteria	
4. Wire Rope Safety Barrier (WRSB)	Tensioned wire ropes mounted on steel posts, flexible type barrier.	Roadside & medians with high accident potential	- Greater deflection (1–2 m) - Absorbs energy with minimal injury - Requires sufficient offset clearance and proper tensioning	
5. Crash Cushions / Impact Attenuators	Energy-absorbing devices placed in front of hazards or barrier ends.	Ramp terminals, bridge piers, toll plazas, gore areas	- Designed to decelerate errant vehicles - Can be sacrificial (crushable) or reusable (rebounding) - Must meet NCHRP 350 or EN 1317 standards	
6. Bridge Parapet (Rigid)	Reinforced concrete wall on the edge of bridges or viaducts.	Edge of bridges or elevated roads	- Minimum height: 1.1 m - Must prevent vehicles from falling - Often integrated with pedestrian handrails	

Introduction to Intelligent Transport Systems (ITS).

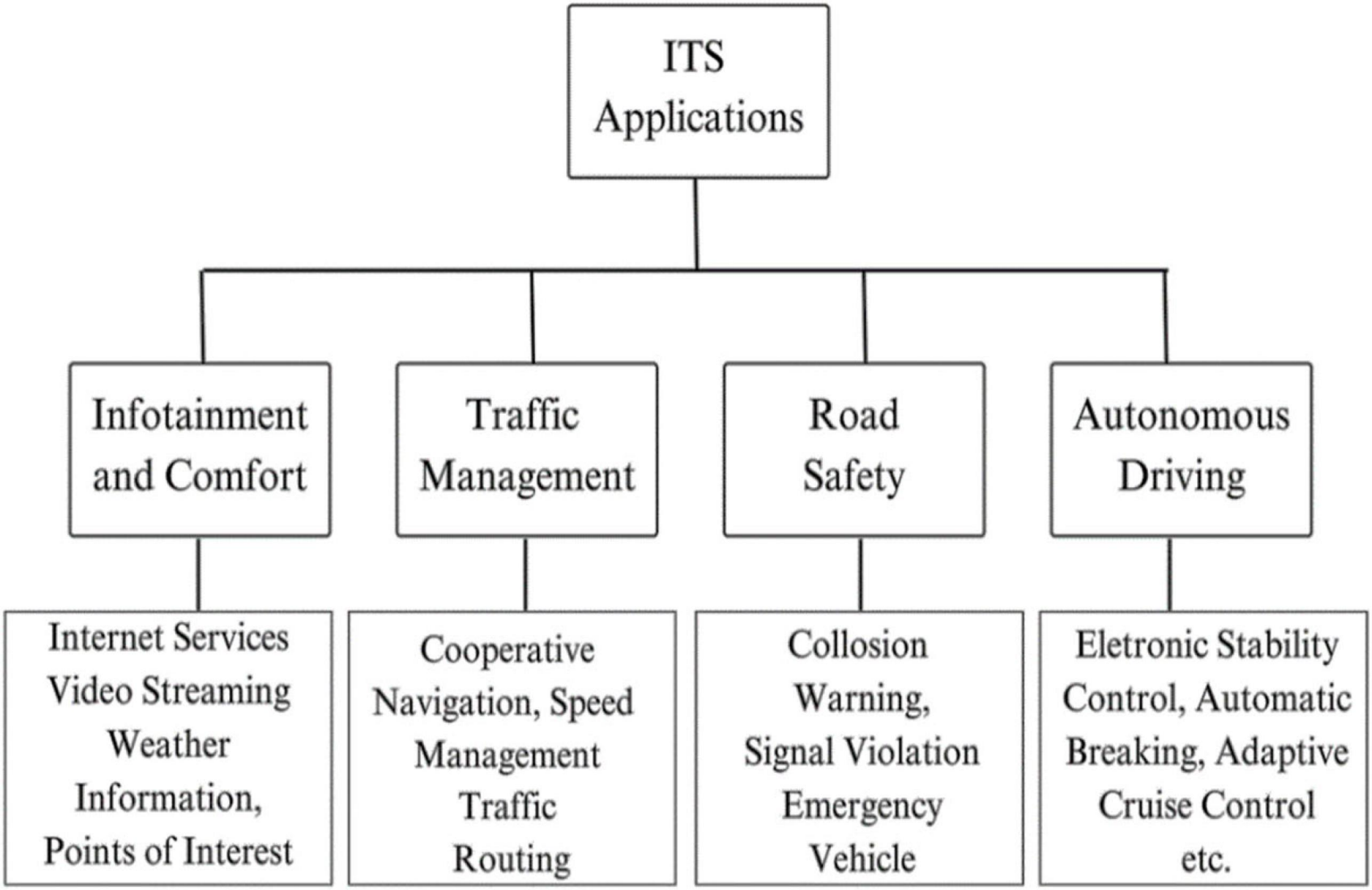
Why is an ITS needed?

N-iX

Source: ScienceDirect

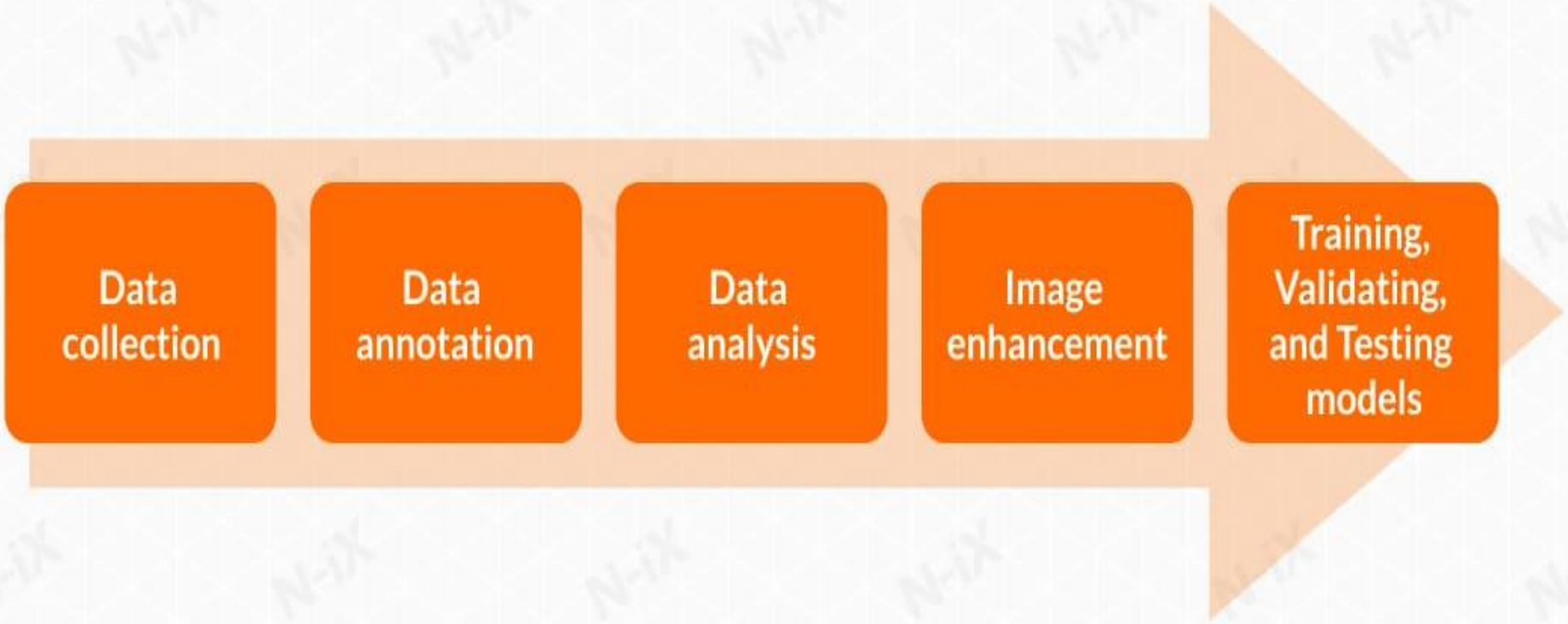


Definition	Application of advanced technologies (ICT, electronics, sensors) in transport systems to enhance safety, efficiency, and sustainability.
Key Objectives	- Improve road safety - Enhance traffic efficiency - Reduce environmental impact - Provide traveler information - Support multimodal transport
Core Technologies Used	- Sensors and detectors - GPS and satellite systems - Wireless communication (V2V, V2I) - Data analytics and AI - Cloud computing
ITS Applications	- Adaptive traffic signals - Electronic toll collection (ETC) - Smart parking - Real-time transit tracking - Driver assistance systems
Benefits	- Reduced congestion - Fewer accidents - Lower emissions - Improved public transport reliability - Better traveler experience
Examples Worldwide	- FASTag in India (ETC) - London Congestion Charging - Japan’s VICS system - USA’s Connected Vehicle Pilot Program
Future Trends	- Autonomous vehicles - 5G-enabled transport networks - Integration with smart cities - AI-driven predictive traffic management

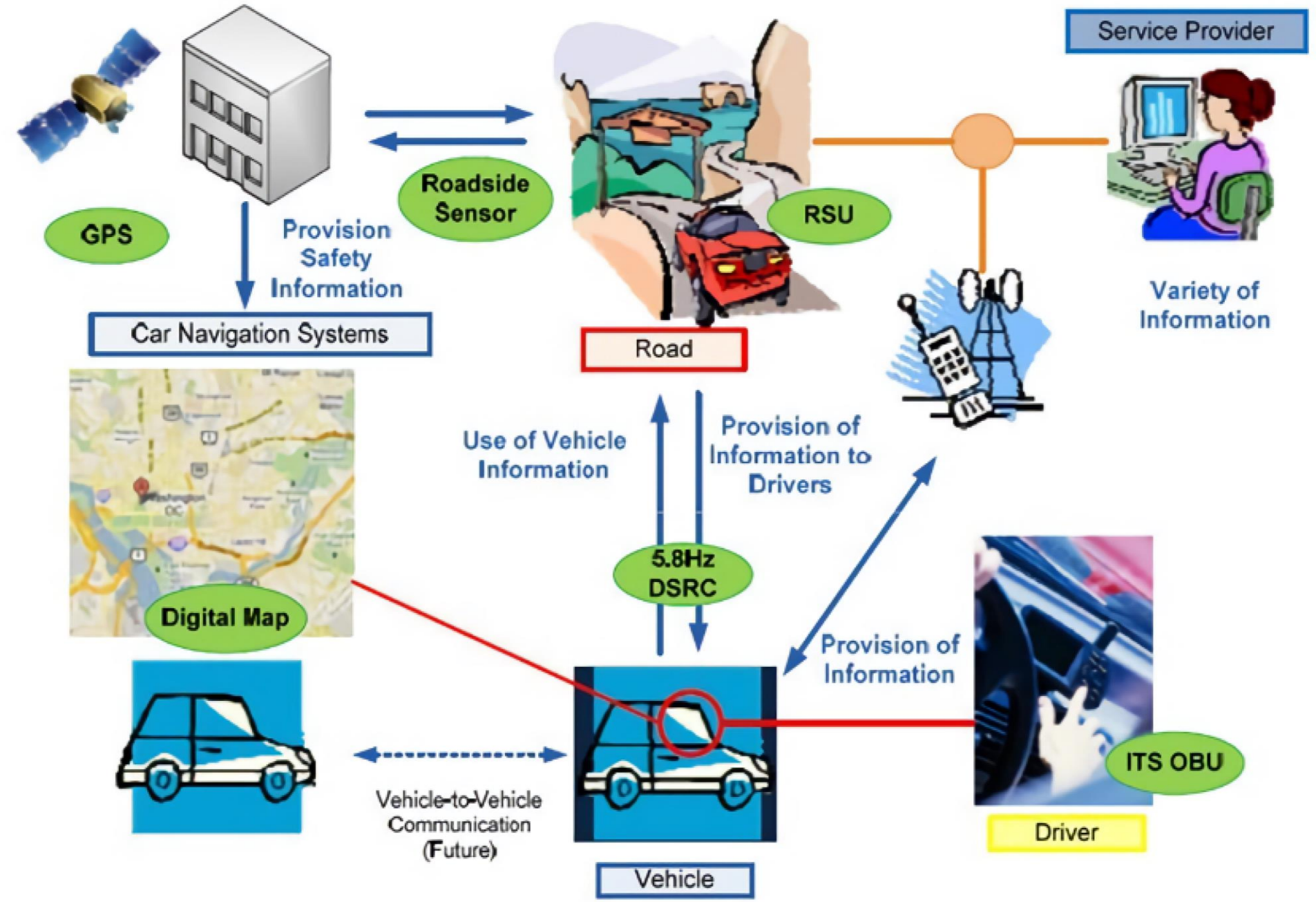

ITS Applications
Infotainment and Comfort
Traffic Management
Road Safety
Autonomous Driving
Internet Services Video Streaming Weather Information, Points of Interest
Cooperative Navigation, Speed Management Traffic Routing
Collosion Warning, Signal Violation Emergency Vehicle
Eletronic Stability Control, Automatic Breaking, Adaptive Cruise Control etc.

Stages of Intelligent Transport System development

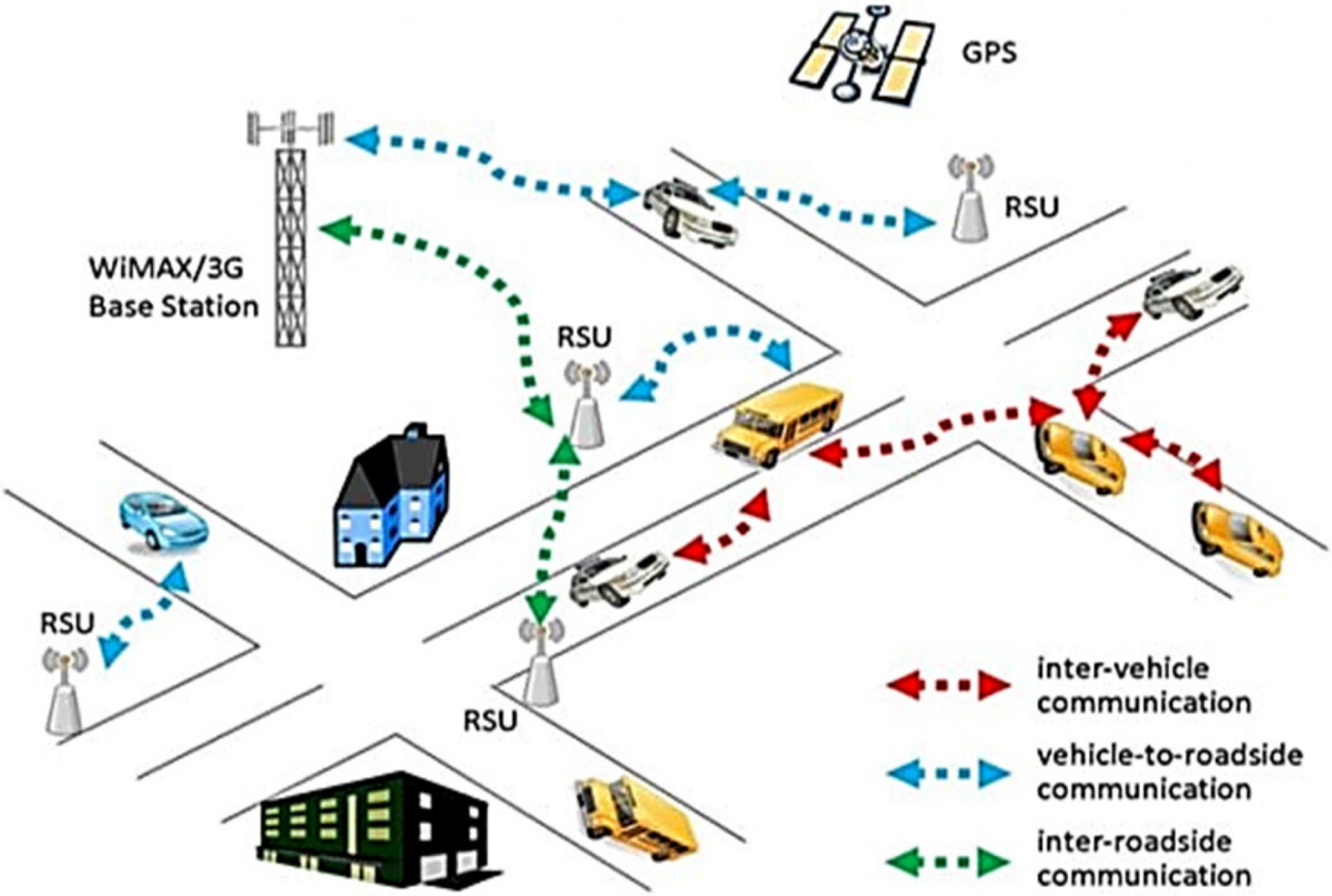
N-iX



ITS communication system



ITS example with GPS



Best Practice	Description	Application in India
Electronic Toll Collection (ETC)	Use of RFID tags and automated systems for toll payments	FASTag implemented nationwide by NHAI
Adaptive Traffic Signal Control	Real-time adjustment of traffic signals based on vehicle flow	Integrated Traffic Management System (ITMS) in cities like Delhi, Bengaluru, Pune
Area-based Congestion Charging	Charging vehicles entering high-congestion zones to reduce traffic	Pilot projects proposed in Mumbai and Delhi
Automatic Number Plate Recognition (ANPR)	Surveillance and enforcement using vehicle number plate detection	Used for e-challan systems in Hyderabad, Delhi, Chennai
Smart Public Transport Management	Real-time tracking, route optimization, and fleet management	Chalo App, Delhi Cluster Buses, MoBus Bhubaneswar
Passenger Information Systems (PIS)	Real-time updates on arrival, departure, and delays	Installed in metro systems (Delhi, Bengaluru), smart bus stops in multiple cities
Intelligent Parking Systems	Smart sensors and apps for real-time parking availability	Pilot smart parking projects in Chandigarh, Bhopal, and Delhi
Emergency Response Systems	GPS tracking and communication for ambulance dispatch	102/108 Ambulance Tracking in Gujarat, Andhra Pradesh
Vehicle Tracking Systems (VTS)	GPS-based tracking of public and freight transport	AIS-140 compliant VTS for public transport mandated by MoRTH
Integrated Command & Control Centres (ICCC)	Centralized monitoring of traffic, transport, and emergency services	Part of Smart Cities Mission (e.g., Bhopal, Surat, Pune)